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# Explicit force-based control strategies for compliant robot-assisted ultrasound imaging

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## Abstract

In this paper, we present a robotic ultrasound system that implements explicit contact force control and force-based full probe orientation optimization to achieve stable, responsive, and high-quality ultrasound imaging. Traditional robotic ultrasound systems often rely on implicit force control methods, such as admittance control, which are limited by their underlying motion-control loops. In contrast, our approach directly regulates contact force and moment at the end-effector, enabling rapid adaptation to heterogeneous tissue properties and dynamic environments. We benchmark the proposed explicit force controller against a state-of-the-art integral adaptive admittance controller, demonstrating a significant reduction in phase lag from 121° to 5.76° at force tracking frequencies exceeding typical respiratory rates. In a second set of benchmarking experiments, compared to the admittance-based method, the explicit force controller achieves a reduction in average force tracking error of 77.7 % when the phantom is moving toward the transducer, and 72.6 % when the phantom is moving away. We integrate the explicit force and moment controller into a six-DoF haptic framework that renders physically-grounded interaction forces to the operator while the robot autonomously regulates contact force and optimizes probe alignment based on acoustic coupling. Validation across static and dynamic scans, as well as under external perturbations, shows that the system consistently maintains target force and moment profiles, aligns the probe with local surface normals, and adapts to changing contact conditions. Experimental results demonstrate that explicit force-based control improves ultrasound image quality, as quantified by confidence maps, compared to a manual haptic scan. These findings support the use of explicit force and moment control as an effective approach for robotic ultrasound imaging.

## Keywords

robot-assisted ultrasound, explicit contact force control, force-based full probe orientation optimization

## 1. Introduction

Ultrasound is among the most widely used diagnostic imaging modalities today, owing to its low cost, inherent safety due to the absence of ionizing radiation, and portability (Szabo, 2004). Consequently, ultrasound is indispensable across diverse healthcare settings, from hospitals to remote clinics, facilitating timely and effective diagnosis of conditions such as cardiac disease, liver pathology, and prenatal health (Moore and Copel, 2011; Picano, 2005). Despite these significant advantages, ultrasound imaging faces a fundamental challenge: acquiring high-quality images requires skilled sonographers trained in the complex manual manipulation of the handheld transducer (Avola et al., 2022). This reliance on specialized human operators creates bottlenecks in healthcare delivery, particularly in low-resource regions where trained personnel are scarce (Gopaul et al., 2018; Stewart et al., 2020).

Furthermore, sonographers suffer from a disproportionately high rate of work-related musculoskeletal disorders caused by repetitive motions, sustained transducer pressure, and awkward postures (Pike et al., 1997). Epidemiological studies indicate that between 76% to 90% of sonographers practice in pain, with up to one in five practitioners sustaining a career-ending injury, further exacerbating workforce shortages and limiting patient access to ultrasound imaging (Al-Rammah et al., 2017; Evans et al., 2009; Zangiabadi et al., 2024).

Robotic ultrasound systems, such as the one presented in Figure 1, offer a compelling solution by assisting human operators in performing ultrasound scans. In robotic

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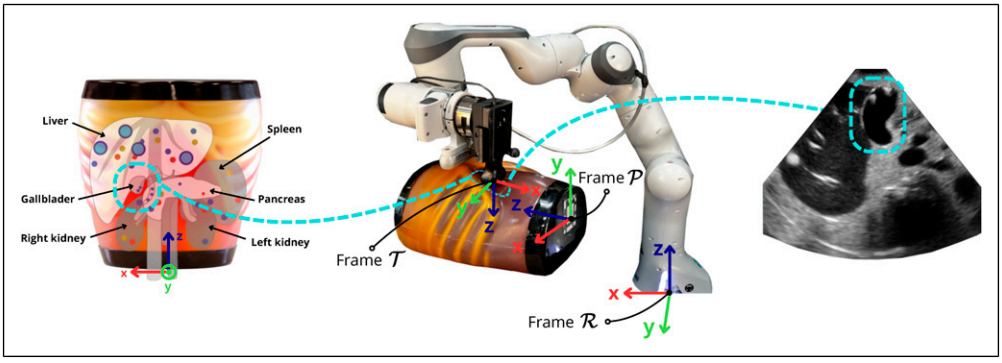



**Figure 1.** Robotic ultrasound acquisition system (with a focus on the gallbladder as an example). **Left**: Abdominal ultrasound phantom with right-anterior-superior coordinate frame and labeled organs. **Middle**: Seven-degree-of-freedom robot manipulating ultrasound transducer over the gallbladder. **Right**: Robotically acquired ultrasound image with the gallbladder highlighted.

ultrasound, the transducer is mounted onto a robotic manipulator capable of precise, repeatable movements, potentially reducing operator fatigue and variability in image quality (Akbari et al., 2021; Kim et al., 2021; Mathur et al., 2019). Such systems can also enhance clinical workflow efficiency and consistency, significantly alleviating the burden on skilled technicians (Jiang et al., 2023).

Moreover, robot-assisted ultrasound platforms facilitate teleoperation, allowing expert clinicians to remotely control the scanning procedure through haptic interfaces (Conti et al., 2014; Piedra et al., 2024). Figure 2 shows such a haptic interface used for remotely performing ultrasound scanning with force and moment feedback rendered to the operator. The remote scanning capability is especially beneficial in underserved or rural settings, extending expert care to areas lacking local specialized medical personnel (Arbeille et al., 2016; Britton et al., 2019; Monsky et al., 2025).

Despite this promise, the clinical utility of existing robotic ultrasound platforms is often constrained by their reliance on compliance-based control methods, such as admittance or impedance, as forms of implicit force regulation (Jiang et al., 2020; Keemink et al., 2018; Ma et al., 2021b). Specifically, admittance control measures the interaction force to command a desired motion, while impedance control measures motion deviations to command a desired force. Both approaches are considered implicit because they rely on motion within the control loop (either as a measurement or a command), preventing a direct and independent regulation of the contact force.

In contrast, explicit force control measures the interaction force and directly commands a corrective force at the actuation level, removing motion from the control loop entirely (Whitney, 1987). Because implicit architectures introduce motion-based dependencies, they can suffer from phase lag and a sluggish response that struggles to maintain accuracy during dynamic contact (Volpe and Khosla, 1993; Whitney, 1987). Crucially, inaccurate force tracking leads to uncontrolled and variable tissue deformation, which compromises the repeatability of image acquisition (Šarabon et al., 2019). Achieving high-precision force regulation is essential for image stability and accurate 3D reconstruction (Huang et al., 2019).

Beyond axial force, optimizing probe orientation is vital for maintaining acoustic coupling and image clarity. Existing strategies typically address this through indirect, motion-based methods such as image-derived velocity commands (Chatelain et al., 2017; Welleweerd et al., 2021). While some systems incorporate limited moment feedback

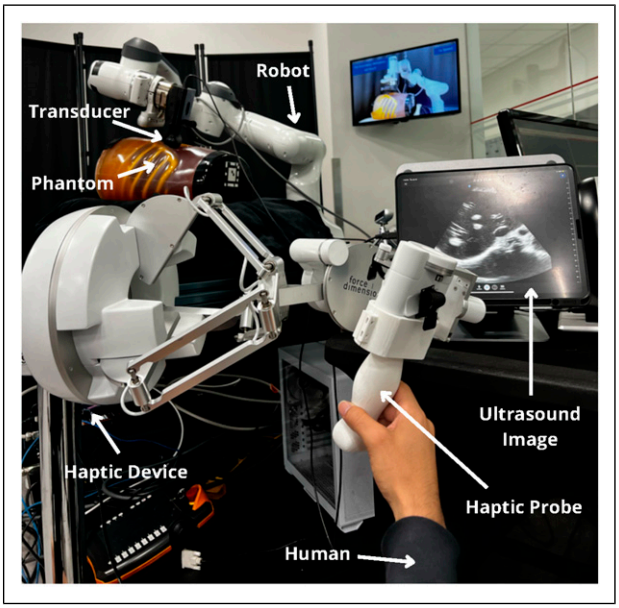



**Figure 2.** Haptically-guided ultrasound setup. **Back**: Robotic manipulator scans a medical phantom using explicit force/moment control. **Front**: Operator uses a six-degree-of-freedom haptic device to guide the robot while monitoring ultrasound images. Force feedback communicates contact pressure, while moment feedback aligns the device with the robot's orientation.

(Huang et al., 2019), explicit moment control remains notably absent in the literature.

In this paper, we present a robotic ultrasound system based on explicit contact force control and force-based orientation optimization. Our approach, derived from the operational space formulation (Khatib, 1987), directly regulates contact forces and moments to ensure stable, responsive, and accurate scanning. To address current limitations in the field, we provide a comparative analysis with a state-of-the-art implicit force controller, highlighting the advantages of explicit force control.

This paper is organized as follows. Section 2 provides an overview of literature related to the use of robotics in the field of ultrasound imaging, with an emphasis on developments in contact force control and full probe orientation optimization. Section 3 introduces a control framework for explicit motion and force control in robotic ultrasound imaging, while Section 4 shows a control framework for haptically-guided robotic ultrasound. Section 5 presents experimental results performed on an abdominal phantom, and Section 6 concludes the paper.

## 2. Related work

The field of robotic ultrasound imaging has recently advanced significantly, driven by the integration of robotics, artificial intelligence, and imaging technologies. These developments address critical limitations in manual ultrasound procedures, such as operator fatigue, inconsistent image quality, and limited access to expert clinicians, particularly in remote or underserved areas. Salcudean et al. (2022) categorize robotic ultrasound systems into teleoperated and autonomous platforms, highlighting the importance of contact force control and probe orientation optimization to achieve effective robotic imaging. Despite these insights, translating these concepts into robust clinical solutions remains challenging.

### 2.1. Contact force control

Ensuring consistent contact force between the ultrasound probe and patient anatomy is essential for stable, high-quality imaging. Traditional approaches often rely on position- and/or velocity-based motion loops to implicitly regulate contact forces (Jiang et al., 2020; Keemink et al., 2018; Ma et al., 2021b). These methods utilize motion within the force control loop, either as the feedback measurement or the control output. Alternatively, indirect methods employ image-derived feedback to adjust the probe pose based on inferred contact, frequently without closing the loop on measured forces (Abolmaesumi et al., 2002; Zettinig et al., 2017; Jiang et al., 2021a, 2021b, 2022).

*2.1.1. Implicit and explicit force control.* A fundamental distinction exists between implicit and explicit force regulation. Implicit methods, such as admittance and impedance control, rely on motion within the control loop. Specifically, admittance control measures the interaction force to command a desired motion, whereas impedance control measures motion deviations to command a desired force (Dyck et al., 2022; Keemink et al., 2018). While both are widely used in robotic applications, impedance control requires knowledge of the environment's position (Suligoj et al., 2021). In clinical settings, the location of the scanning surface changes due to patient respiration and shifting.

In contrast, admittance control measures interaction forces to command a desired motion, allowing it to track a desired force without requiring knowledge of the environmental position (Keemink et al., 2018; Lin et al., 2025). Recent advancements have sought to address the limitations of standard admittance control in soft, uncertain environments, such as those encountered in robotic ultrasound imaging. Specifically, integral adaptive admittance control (IAAC) represents the current state of the art, utilizing integral action to eliminate steady-state force errors and adaptive gains to estimate unknown tissue stiffness in real time (Jiang et al., 2024). However, admittance-based force control, as with any compliance-based approach, is subject to "sluggish" performance compared to explicit methods because implicit methods regulate force through an inner motion-control loop (Whitney, 1987).

Explicit force control measures the interaction force and directly commands a corrective force at the actuation level, removing the intermediate motion loop entirely (Whitney, 1987). Controlling force at the end-effector, as done in both impedance-based and explicit force control, is most effectively achieved through closed-loop torque control on robotic manipulators equipped with joint torque sensors (Dyck et al., 2022). This is typically implemented using operational space control frameworks, which utilize task-space inertia matrix decoupling to enhance stability and responsiveness during dynamic interactions (Khatib, 1987; Welleweerd et al., 2021). While various systems have incorporated force feedback for incremental position-based force regulation (Fang et al., 2017; Huang et al., 2018, 2019; Jiang et al., 2021; Ma et al., 2021; Ning et al., 2021a, 2021b; Wang et al., 2023), explicit contact force control remains relatively unexplored in the field of robotic ultrasound.

*2.1.2. Performance and stability considerations.* The stability of any force control scheme is highly sensitive to the interaction environment. Whitney (1977) noted that admittance control stability depends on the product of terms including the sampling period, controller gains, and environmental stiffness. Proportional explicit force control can also become unstable when interacting with rigid surfaces, whereas stability typically improves with increased environmental compliance (An and Hollerbach, 1987). The stability boundary is defined by the interaction between the robot, the sensor, and the workpiece (Eppinger and Seering, 1987a).

While extremely compliant surfaces can lead to a loss of force feedback, high-stiffness environments can drive the system toward instability. Volpe and Khosla (1993) demonstrated that position-based force control inherently adds stiffness to the plant, which can be fundamentally destabilizing. Furthermore, performance is strictly bounded by hardware constraints, including actuator bandwidth, sensor filtering, and digital sampling rates (Eppinger and Seering, 1987b; Lawrence, 1988).

*2.1.3. Image-driven force optimization.* Beyond the force control architecture, a critical challenge lies in determining the optimal contact force for different anatomical targets. While most systems utilize a constant reference force, recent advancements have introduced artificial intelligence-driven strategies to dynamically derive desired force values from real-time ultrasound images (Zakeri et al., 2025).

*2.1.4. Clinical force magnitudes.* Robotic ultrasound controllers must operate within specific force thresholds that vary significantly across clinical specialties (Suchoń et al., 2025). For abdominal procedures, baseline studies have quantified average axial forces at 7.0 N (Gilbertson and Anthony, 2013) and (8.2 ± 4.3) N, with a range of 1.2 N to 36.5 N (Dhyani et al., 2017). Patient-specific factors further influence these requirements; for instance, scanning high-BMI subjects requires higher mean forces (10.5 N) compared to normal-BMI subjects (7.9 N) to maintain diagnostic quality (Dhyani et al., 2017). Maintaining precise regulation is essential because force variations as small as 1.5 N significantly degrade measurement repeatability (Šarabon et al., 2019), and 5.0 N is sufficient to almost completely compress vascular structures (Gilbertson and Anthony, 2015).

The mechanical interaction between the probe and underlying tissue serves as the basis for ultrasound elastography, which controls deformation to characterize tissue stiffness as a biomarker for pathology (Ozturk et al., 2018). Recent advancements leverage deep learning for sound speed inversion from channel data, providing diagnostic capabilities similar to shear wave imaging while improving frame rates and robustness against motion artifacts (Feigin et al., 2020).

*2.1.5. Environmental contact estimation.* Accurate and real-time estimation of environmental contact geometry significantly enhances robotic force control performance. Adaptive stiffness control methods adjust gains dynamically based on interaction dynamics, typically reducing stiffness and increasing damping in compliant directions (Hennersperger et al., 2017; Jiang et al., 2023; Salisbury, 1980). Jorda et al. (2022) introduced a probabilistic framework using a force-space particle filter to estimate uncertain environmental contact states, improving the system's ability to adapt to changing interactions.

Existing systems with force feedback have used adaptive control strategies to modify control parameters online. Cortesão et al. (2006) developed a method to change controller stiffness based on the environment, employing higher force control gains when the environment is more compliant to maintain responsiveness, and reducing said gains for stiffer environments to maintain stability. This approach has been successfully applied to ultrasound scanning on human subjects, where it accounts for the variability in tissue stiffness across different anatomical regions (Santos and Cortesão, 2018).

## 2.2. Full probe orientation optimization

Optimizing the ultrasound probe orientation is essential for maximizing acoustic coupling, improving image clarity, and reducing artifacts. Effective probe alignment typically involves orienting the probe normal to the tissue surface (Ihnatsenka and Boezaart, 2010).

One common approach relies on image-derived feedback, where image quality metrics are translated into angular velocity commands to maintain contact (Chatelain et al., 2017). Alternatively, deep learning has been used to estimate the surface normal angle directly from 3D ultrasound volumes, enabling immediate orientation corrections (Osburg et al., 2022). Other systems utilize external vision; for instance, RGB-D cameras can recognize anatomical targets to compute an orientation perpendicular to the skin (Ma et al., 2021b), while others estimate the normal by analyzing the 3D curvature of pre-planned sweep trajectories (Jiang et al., 2021b). Distance-based sensing offers a robust alternative, exemplified by the A-SEE system, which uses four laser distance sensors embedded in the end-effector for autonomous self-positioning without pre-operative data (Ma et al., 2022). Finally, hybrid methods use parallel load cells to provide limited moment feedback for fine-tuning the orientation from an initial vision-based estimate (Huang et al., 2019).

Despite these advancements, many probe orientation methods rely on admittance controllers that convert measured errors into motion commands. These controllers suffer from the same latency as admittance-based contact force control. To achieve more responsive alignment, Khansari et al. (2016) demonstrated a zero-moment compliant strategy that minimizes measured moments about the contact point to align surfaces. Adapting this to ultrasound can help the probe remain flush with the tissue by driving misalignment moments to zero. However, explicit closed-loop moment control for optimizing ultrasound probe orientation remains unexplored in the literature.

## 2.3. Evaluating ultrasound image quality

The objective evaluation of ultrasound image quality is a multi-layered challenge, spanning from low-level signal

integrity to high-level clinical utility. Quantifying this quality is essential for robotic systems to provide closed-loop feedback; early frameworks utilized quantitative interfaces to statistically assess penetration depth and resolution as indicators of diagnostic effectiveness (Hemmsen et al., 2010).

*2.3.1. Signal-based metrics and noise characteristics.* Traditional assessment often utilizes full-reference metrics like peak signal-to-noise ratio and mean squared error (MSE). However, these metrics frequently fail to capture features relevant to clinical diagnosis because they do not account for the unique physics of ultrasound (Chow and Paramesran, 2016). Specifically, ultrasound images are characterized by multiplicative speckle noise, which is typically modeled by a Rayleigh distribution for envelope-detected signals or a Gamma distribution for intensity-based data (Loizou et al., 2006). While the structural similarity index is more consistent with human visual perception than MSE, it remains sensitive to the image translations and rotations inherent in freehand scanning, limiting its utility for real-time robotic guidance (Rangaraju et al., 2012).

*2.3.2. Computing confidence maps.* To address the lack of absolute reference images in clinical settings, confidence maps provide a no-reference estimate of uncertainty caused by acoustic attenuation and shadowing (Karamalis et al., 2012). These maps utilize a random walk framework to model the probability of sound wave propagation, identifying regions with poor acoustic coupling or significant signal loss. The resulting per-pixel confidence scores serve as a robust feedback signal for robotic control; for instance, several studies have utilized confidence map moments to design visual servoing laws that optimize probe orientation in real time (Chatelain et al., 2015, 2017). More recently, this approach has been extended to autonomous breast scanning, where confidence-driven control keeps the scanning trajectory tangent to the tissue surface without requiring a prior patient model (Welleweerd et al., 2021).

*2.3.3. Content-aware quality assessment.* Despite their utility in identifying acoustic coupling, confidence maps primarily evaluate pixel-level signal strength without considering the semantic significance of the image content (Song et al., 2022). A high-confidence image may still be diagnostically useless if it fails to capture the required anatomical plane. To bridge this gap, deep learning-based quality assessment frameworks have been developed to learn evaluation standards directly from clinical experts. For example, systems for fetal ultrasound assess quality based on the clear depiction of key structures, such as the stomach bubble or umbilical vein (Wu et al., 2017). Recent robotic architectures have integrated bilinear convolutional neural networks to achieve high correlation with physician scoring, enabling autonomous systems to distinguish between noise and true diagnostic information (Song et al., 2022).

## 2.4. Teleoperated ultrasound systems

Teleoperated robotic systems expand clinical reach by allowing expert sonographers to perform examinations from remote locations. These platforms typically utilize haptic interfaces to map clinician hand movements to a remote manipulator while providing tactile feedback of probe-patient contact (Conti et al., 2014). Our previous work developed a framework for recording and replaying expert scanning strategies, enabling autonomous survey scans that can be followed by detailed remote-guided examinations (Piedra et al., 2024). Despite these capabilities, achieving high-fidelity remote interaction requires addressing the technical challenges of communication latency and bandwidth.

To ensure stability over long distances, architectures such as the dual-proxy model provide local autonomy that maintains consistent contact forces despite signal delays (Jorda et al., 2022). Augmented reality and predictive force models further assist operators by projecting anatomical models and predicting contact forces to compensate for time-lagged visual feedback (Fu et al., 2023). Collaborative methods have also incorporated shared control, where the robot autonomously manages low-level stabilization while the clinician directs the high-level diagnostic trajectory (Si et al., 2024).

Although the feasibility of teleoperated robot-assisted ultrasound has been demonstrated, widespread clinical transition depends on leveraging high-speed networks to maintain image feedback while refining six-degree-of-freedom (six-DoF) haptic interfaces to improve sonographer control and diagnostic reproducibility across standardized commercial platforms (Hidalgo et al., 2023).

## 2.5. Clinical adaptability and standardization

Widespread adoption of robotic ultrasound hinges on transitioning from experimental feasibility to standardized, reproducible diagnostic capabilities. Historically, Ossoinig (1979) demonstrated that standardization of instrumentation and techniques in echography enables highly reproducible tissue differentiation. Modern robotic systems offer a pathway to re-introduce this rigor by minimizing the inter-operator variability inherent in manual ultrasound scanning; for instance, Kojcev et al. (2017) showed that robotic acquisitions significantly improve trajectory and coverage reproducibility compared to expert sonographers. Furthermore, stable robotic control is helpful for accurate 3D volume reconstruction; Janvier et al. (2008) demonstrated that robotic 3D ultrasound allows for sub-millimeter positioning accuracy (0.46 mm to 0.75 mm mean error) and

quantification errors of −0.9% to 3.0% for vascular stenoses.

As systems evolve toward supervised autonomy, establishing rigorous performance benchmarks and safety standards becomes critical for commercialization (Haidegger, 2019). To this end, Nagy and Haidegger (2022) proposed capability assessment models to quantify automated subtask performance, such as force application, against benchmarks. Future research should focus on integrating these standardized protocols with advanced interfaces, such as immersive virtual reality (Shyam et al., 2023), to help systems adapt to diverse patient anatomies while maintaining clinical safety.

## 3. Explicit motion and force control for robotic ultrasound imaging

We present a robotic ultrasound acquisition system composed of an ultrasound transducer and a compliant, seven-DoF torque-controlled robotic manipulator equipped with a force/torque sensor at the wrist. These components and the ultrasound phantom, along with its labeled anatomy and right-anterior-superior (RAS) anatomical coordinate frame, are shown in Figure 1.

The manipulator's kinematic structure satisfies the six-DoF requirement for full probe pose control while providing a redundant degree of freedom that enables control of the robot's posture, analogous to the elbow positioning of a human arm. The redundancy allows the manipulator to perform motions within the null-space of the probe's task-space control. Moreover, the manipulator's integrated joint torque sensors are fundamental to our explicit force control approach, providing the high-frequency feedback necessary for controlling force at the end-effector. The ultrasound transducer is rigidly attached to the force/torque sensor, which is rigidly attached to the robotic manipulator's wrist. All data from the wrist-mounted force/torque sensor is dynamically compensated for the mass and inertia of the ultrasound probe and its coupling hardware.

### 3.1. Operational space formulation

We adopt the operational space control framework introduced by Khatib (1987) to describe the robot's dynamics at the point, $\mathcal{O}$, corresponding to the center of the transducer lens. For a robot with $n$ generalized degrees of freedom $q \in \mathbb{R}^n$, the joint-space dynamics are

$$A(q)\ddot{q} + b(q,\dot{q})\dot{q} + g(q) = \tau, \tag{1}$$

where $\dot{q}$ and $\ddot{q}$ are the joint velocities and accelerations, respectively, $A(q) \in \mathbb{R}^{n\times n}$ is the joint-space inertia matrix, $b(q,\dot{q})$ is the vector of Coriolis and centrifugal forces, $g(q)$ is the gravity vector, and $\tau \in \mathbb{R}^n$ is the vector of joint torques. Note that $A(q)$ is a symmetric positive-definite matrix.

Let $x \in \mathbb{R}^m$ denote the operational (task) space coordinates associated with the operational point $\mathcal{O}$. The task Jacobian $J(q) \in \mathbb{R}^{m\times n}$ relates joint-space velocities to task-space velocities through

$$\dot{x} = J(q)\dot{q}. \tag{2}$$

Since the robot manipulator is redundant with respect to the task of controlling the transducer six-dimensional motion and force, the redundancy is controlled with a joint configuration-holding control torque vector $\tau_0 \in \mathbb{R}^n$ projected onto the dynamically-consistent task null-space, $N(q)$. The joint-space control torques are

$$\tau = J^T(q)F + N^T(q)\tau_0, \tag{3}$$

where $N(q) = I_{7\times 7} - \overline{J}(q)J(q)$.

To express the robot's dynamics in operational space, we use the dynamically-consistent generalized inverse of the task Jacobian, $\overline{J}(q)$, which is the solution to the generalized inverse of $J(q)$ that minimizes the robot's instantaneous kinetic energy. The dynamically-consistent inverse is defined as

$$\overline{J}(q) = A^{-1}(q)J^T(q)\left(J(q)A^{-1}(q)J^T(q)\right)^{-1}. \tag{4}$$

Multiplying equation (1) by $\overline{J}^T(q)$ yields the task-space dynamics

$$\begin{aligned} &\overline{J}^T(q)\left(A(q)\ddot{q} + b(q,\dot{q})\dot{q} + g(q) = \tau\right), \\ &\Rightarrow \Lambda(x)\ddot{x} + \mu(x,\dot{x}) + p(x) = F, \end{aligned} \tag{5}$$

where $\Lambda(x) \in \mathbb{R}^{m\times m}$ is the task-space inertia matrix at $\mathcal{O}$, $\mu(x,\dot{x}) \in \mathbb{R}^m$ is the vector of Coriolis and centrifugal forces in task space, $p(x) \in \mathbb{R}^m$ is the task-space gravity vector, and $F \in \mathbb{R}^m$ is the generalized task force. Explicitly, these quantities are given by

$$\Lambda(x) = \left(J(q)A^{-1}(q)J^T(q)\right)^{-1}, \tag{6}$$

$$\mu(x,\dot{x}) = \overline{J}^T(q)b(q,\dot{q}) - \Lambda(q)\dot{J}(q)\dot{q}, \tag{7}$$

$$p(x) = \overline{J}^T(q)g(q). \tag{8}$$

In general, since the joint-space inertia matrix $A(q)$ is positive-definite, it follows that $\Lambda^{-1}(x)$ is positive semi-definite. Specifically, equation (5) holds for non-singular joint configurations $q$ because the task Jacobian $J(q)$ is full rank, $\Lambda^{-1}(x)$ is positive-definite, and so $\Lambda(x)$ is well-defined and also positive-definite. Near singular joint configurations, $J(q)$ becomes rank-deficient, requiring special operational space control strategies.

In the context of ultrasound imaging, the frame defined at the operational point $\mathcal{O}$ consists of both three-

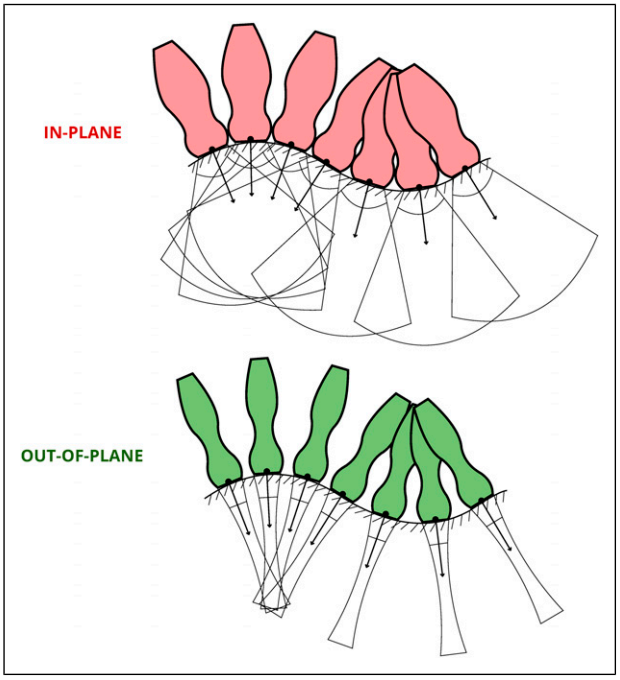



**Figure 3.** In-plane and out-of-plane orientation optimization.

dimensional translational and three-dimensional rotational components. We refer to the translational and rotational components in equation (5) with $(\cdot)_t$ and $(\cdot)_r$, respectively. Using this notation, the center of the transducer lens' position and orientation are represented by $x_t$ and $x_r$. The end-effector's instantaneous linear and angular velocity are $v$ and $\omega$. These quantities are represented by

$$x = \begin{bmatrix} x_t \\ x_r \end{bmatrix}, \qquad \dot{x} = \begin{bmatrix} v \\ \omega \end{bmatrix}. \tag{9}$$

The task forces and moments are denoted by $F$. In this paper, we represent the end-effector's orientation as a rotation matrix $R \in \mathbb{SO}(3)$. The end-effector rotation in task space, $x_r$, is derived from $R$ according to the selected task representation.

### 3.2. Constrained operational space control

Consider a robot manipulator holding an ultrasound probe that is in contact with the environment at $\mathcal{O}$. Now, the robot's task-space dynamics are given by adding the contact force $F_c$ between the probe and the environment to the left-hand side of equation (5) as shown:

$$\Lambda(x)\ddot{x} + \mu(x, \dot{x}) + p(x) + F_c = F. \tag{10}$$

In contact, the task workspace is separated into the force space (directions of constraints) and motion space (directions of free motion). Linearly, the force and motion spaces are defined by orthogonal projection matrices $\Omega_{f,t}$ and $\Omega_{m,t}$, respectively, such that $\Omega_{f,t} + \Omega_{m,t} = I_{3\times3}$. Similarly, in rotation, the force and motion spaces are defined by orthogonal projection matrices $\Omega_{f,r}$ and $\Omega_{m,r}$, respectively, such that $\Omega_{f,r} + \Omega_{m,r} = I_{3\times3}$. The resulting six-dimensional force-space and motion-space projections are given by the following block diagonal matrices, hereafter referred to as the task specification matrices:

$$\Omega_f = \begin{bmatrix} \Omega_{f,t} & 0_{3\times3} \\ 0_{3\times3} & \Omega_{f,r} \end{bmatrix}, \tag{11}$$

$$\Omega_m = \begin{bmatrix} \Omega_{m,t} & 0_{3\times3} \\ 0_{3\times3} & \Omega_{m,r} \end{bmatrix}. \tag{12}$$

We use the unified force and motion controller in operational space to locally control both the constrained motion and force behavior on the robot (Khatib, 1987). With this decoupling technique and assuming perfect estimates of the dynamic components, the motion and force control loops regulate the desired behavior of the decoupled end-effector equivalent to a unit mass system in motion space and a force source in force space. This unified control strategy provides good performance in controlling both the motion of the transducer and its contact with the phantom.

### 3.3. Force-based full orientation optimization

Maintaining zero-moment control at the tip of the ultrasound probe is critical, as any residual moment results in orientation error, which in turn affects the quality and consistency of the reconstructed ultrasound image. Furthermore, moment control must be enforced in both the in-plane and out-of-plane directions for full probe orientation optimization. Figure 3 illustrates the imaging range of the probe in both orientations.

The control force (moment) in rotational force (moment) space, denoted $F_{f,r}$, will be referred to as $M$ throughout the remainder of this paper. The control moment, $M$, is computed to track the desired end-effector moment, $M_d$, using a proportional-integral controller with feedforward moment compensation and velocity-based damping, as defined in the following:

$$M = \Omega_{f,r}\left(M_d - K_{p,f,r}(M_s - M_d) - K_{i,f,r}\int (M_s - M_d)\right). \tag{13}$$

The moment-space proportional and integral gains are $K_{p,f,r}$ and $K_{i,f,r}$. The sensed task moment, $M_s$, is used as feedback to the controller.

Figure 4 illustrates the origin of the moment error in the system. When the robot establishes contact with the

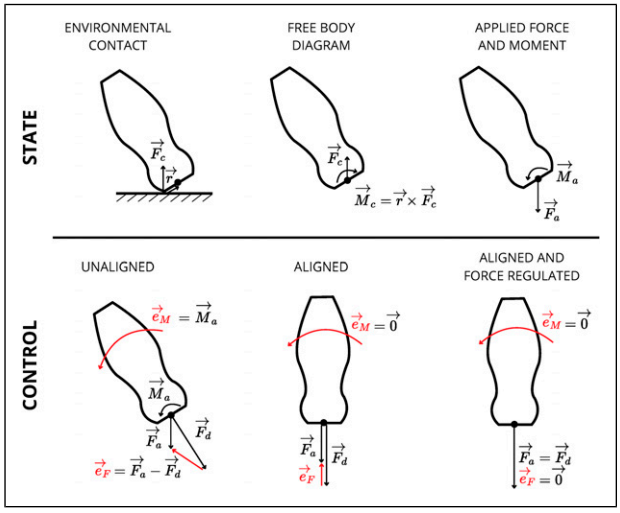



**Figure 4.** Contact state and control definitions.

environment, it experiences a contact force $F_c$, which generates a contact moment $M_c$ about the control point due to the lever arm $r$. This moment represents the effect of the contact force applied by the probe onto the environment.

We assume that the contact force $F_c$ is accurately measured by the force/torque sensor and is thus equal to the sensed force $F_s$. And therefore we also assume that the contact moment $M_c$ is equal the sensed moment $M_s$.

The control objective is to drive the applied moment at the control point to zero. Accordingly, we define the moment error as the difference between the applied moment and the desired moment (set as zero), such that

$$\begin{aligned} e_M &= M_a - \cancelto{0}{M_d}, \\ &= M_a. \end{aligned} \tag{14}$$

The controller minimizes this error, causing the probe to rotate until the moment is eliminated. The zero-moment goal is achieved when the transducer lens is aligned with the environment.

### 3.4. Contact force control

The control force in translational force space, $F_{f,t}$, is computed to follow the desired end-effector force, $F_d$, through a proportional-integral controller with feedforward force compensation as follows:

$$\begin{aligned} F_{f,t} = \Omega_{f,t} \Big( & F_d - K_{p,f,t}(F_s - F_d) \\ & - K_{i,f,t} \int (F_s - F_d) \Big). \end{aligned} \tag{15}$$

The translational force-space proportional and integral gains are $K_{p,f,t}$ and $K_{i,f,t}$. The sensed linear force, $F_s$, is used as feedback to the controller.

The control objective, illustrated in Figure 4, is to regulate the applied force to the environment, $F_a$, to match the desired force, $F_d$, by minimizing the force error defined as $e_F = F_a - F_d$. It is assumed that the contact force is accurately measured by the force/torque sensor, that is, $F_a = F_s$ and $e_F = F_s - F_d$. Force regulation is successfully achieved as $e_F$ converges to zero, ensuring the applied force tracks the desired value.

### 3.5. Force space damping

The control force in the full force space, denoted $F_f$, is computed by adding velocity-based damping to the concatenated translational and rotational force controllers, $F_{f,t}$ and $F_{f,r}$, as follows:

$$\begin{aligned} F_f &= \begin{bmatrix} F_{f,t} \\ F_{f,r} \end{bmatrix} - \widehat{\Lambda} \Omega_f K_{v,f} \begin{bmatrix} v \\ \omega \end{bmatrix}, \\ &= \begin{bmatrix} F_{f,t} \\ M \end{bmatrix} - \widehat{\Lambda} \Omega_f K_{v,f} \dot{x}. \end{aligned} \tag{16}$$

The estimate of the task-space inertia matrix, $\widehat{\Lambda}$, is used to dynamically decouple the force-space damping. The derivative gain matrix for force-space damping is $K_{v,f}$.

### 3.6. Motion control

Let the desired end-effector position and rotation in task coordinates be denoted $x_{d,t}$ and $x_{d,r}$, respectively. The desired end-effector rotation matrix, $R_d$, is used to derive $x_{d,r}$. The instantaneous angular error corresponding to the error between the actual end-effector rotation and its desired rotation is expressed as

$$\delta\phi = E_r^+ (x_r - x_{d,r}), \tag{17}$$

where the matrix $E_r$ maps the basic Jacobian to the task Jacobian, $J = E_r J_0$, as a function of the task representation, and $E_r^+$ is the pseudoinverse of $E_r$. The end-effector error in motion is then

$$e_m = \begin{bmatrix} e_{m,t} \\ e_{m,r} \end{bmatrix} = \begin{bmatrix} x_t - x_{d,t} \\ \delta\phi \end{bmatrix}. \tag{18}$$

The control force in motion space, $F_m$, is generated by an operational space proportional-derivative controller given by

$$F_m = \widehat{\Lambda} \Omega_m \left( -K_{p,m} e_m - K_{v,m} \dot{x} \right), \tag{19}$$

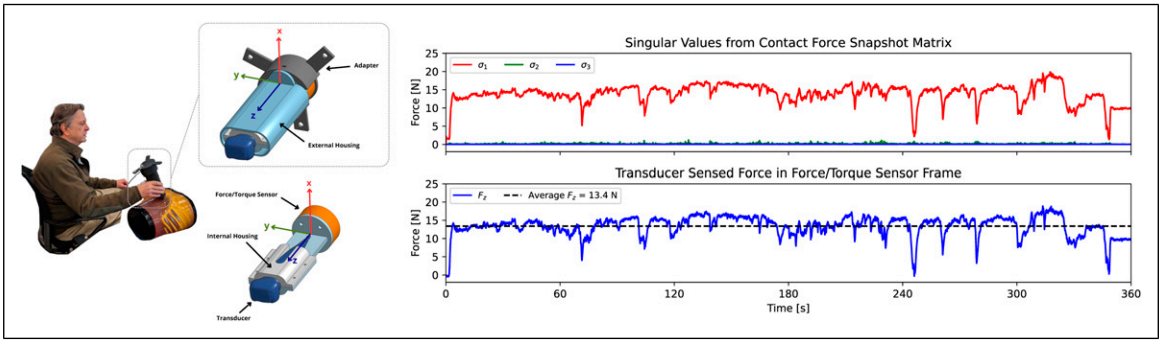



**Figure 5. Left**: Freehand abdominal ultrasound scan setup. **Top right**: Singular values of the contact force snapshot matrix from the full freehand scan, showing that the expert primarily controls force along a single direction. **Bottom right**: Sensed force along the transducer's long axis, $\hat{z}$, in the force/torque sensor frame. The average axial force, $F_z$, is 13.4 N.

The estimate of the task-space inertia matrix, $\widehat{\Lambda}$, is used to dynamically decouple the system. The proportional and derivative gains for motion control are $K_{p,m}$ and $K_{v,m}$.

## 3.7. Unified force and motion control

The control forces in force and motion space are composed together with two additional gravity and Coriolis/centrifugal compensation terms. The result is the dynamically decoupled and unified force and motion-control force, $F$, given by

$$F = F_f + F_m + \widehat{\mu} + \widehat{p}, \tag{20}$$

where $\widehat{\mu}$ is the estimated vector of task-space Coriolis and centrifugal forces, and $\widehat{p}$ is the estimated task-space gravity. Transitions from free-space motion to closed-loop force control are stabilized with a passivity observer (Jorda et al., 2017). The control torques specified to the robot are given by using $F$ with equation (3).

## 3.8. Task specification

*3.8.1. Expert-derived contact force specification.* To establish a clinically relevant desired force, we analyzed an expert radiologist's scanning behavior using the recording mechanism developed in our previous work Piedra et al. (2024). As shown in Figure 5, this hardware captures the transducer's trajectory and applied contact forces during freehand scanning. Using this system, the radiologist performed a comprehensive abdominal scan.

To analyze the recorded force data, we utilized proper orthogonal decomposition (POD), a data-driven technique used to obtain low-dimensional descriptions of high-dimensional data by extracting the principal directions of variance (Chatterjee, 2000). In abdominal ultrasound, multiple contact directions are uncommon due to the use of ultrasound gel, which minimizes friction and lateral forces. Consequently, we assume the radiologist applies force along a single dominant direction. By applying POD to the force snapshot matrix, we evaluate this assumption by examining the resulting singular values.

The three singular values of the force snapshot matrix computed at each time step are shown in Figure 5. The average ratio between the largest and second-largest singular values was 217%, while the average ratio between the largest and smallest singular values was 721%. This significant disparity confirms that the interaction is dominated by a single principal axis. Therefore, we consider the force to be applied in a single direction, which we choose as the transducer's long axis, $\hat{z}$, in all subsequent experiments. A second plot of the sensed force at the probe tip shows an average $F_z$ of 13.4 N, justifying the use of this desired force in our experiments.

*3.8.2. Task specification matrices.* When the end-effector is in contact with the phantom, the force-space projection matrix in translation is

$$\Omega_{f,t} = \widehat{z}\widehat{z}^T, \tag{21}$$

where $\widehat{z}$ is the transducer's long axis and the motion-space projection matrix in translation is $\Omega_{m,t} = I_{3\times3} - \widehat{z}\widehat{z}^T$.

For force-based full orientation optimization, the zero-moment controller is applied in the rotational force space, which is orthogonal to the translational force space. This applies zero-moment control in the transducer lens plane, allowing the transducer lens to conform to the phantom surface while the linear force is controlled along the probe's long axis. The force-space projection in rotation is then

$$\begin{aligned} \Omega_{f,r} &= I_{3\times3} - \Omega_{f,t}, \\ &= I_{3\times3} - \widehat{z}\widehat{z}^T, \end{aligned} \tag{22}$$

and the motion-space projection matrix in rotation is $\Omega_{m,r} = I_{3\times3} - \Omega_{f,r} = \Omega_{f,t}$. This enables control of the

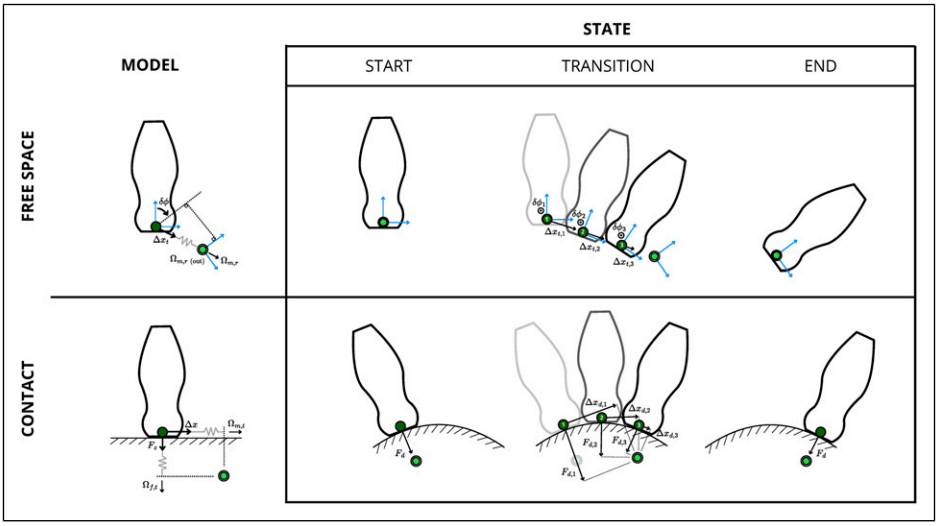



**Figure 6.** Robot proxy (light green) defining desired transducer force and motion at control point (dark green). **Top**: Transducer translating and rotating in free space. **Bottom**: Transducer moving along curved surface while maintaining contact force.

transducer rotation about its long axis, which selects the imaging plane (e.g., transverse, sagittal, oblique) depending on the transducer's orientation relative to the patient or phantom. In practice, this relative orientation is monitored through the camera-based registration presented in Section 5.1.2.

Finally, we have the following task specification matrices (with the matrix sizes removed):

$$\Omega_f = \begin{bmatrix} \hat{z}\hat{z}^T & 0 \\ 0 & I - \hat{z}\hat{z}^T \end{bmatrix}, \quad \Omega_m = \begin{bmatrix} I - \hat{z}\hat{z}^T & 0 \\ 0 & \hat{z}\hat{z}^T \end{bmatrix}. \quad (23)$$

In free space, the robot's task specification matrices are set for pure motion control in translation and rotation, with $\Omega_m = I_{6\times6}$ and $\Omega_f = 0_{6\times6}$.

*3.8.3. Robot proxy.* In general, the robot end-effector goal pose is abstracted as the robot proxy. For clarity, only the translational components are discussed, but this concept naturally extends to rotational control. In motion space, the linear displacement between the robot end-effector position $x_t$ and the robot proxy position $x_p$ defines the desired end-effector translation as

$$x_{d,t} = x_{p,t} \quad (24)$$

In force space, the desired linear force can be specified as the linear displacement between the robot position and its proxy position with a virtual spring,

$$F_d = k_{vir}(x_t - x_{p,t}) \quad (25)$$

where $k_{vir}$ is the spring stiffness. The control forces expressed in both motion and force space are illustrated in Figure 6.

## 4. Haptically-guided robotic ultrasound

Haptic guidance in robot-assisted ultrasound enables clinicians to interactively refine scans, particularly in regions requiring detailed assessment beyond autonomous capabilities. This interaction emulates freehand scanning at a distance, allowing intuitive control without direct contact.

Our framework necessitates a fully actuated 6-DoF haptic interface to support the explicit control of the probe's full pose. In this setup, linear force feedback communicates real-life contact forces to the operator, providing the tactile sensitivity needed to assess tissue stiffness or maintain specific pressure. Furthermore, the inclusion of moment feedback allows the haptic device to align with the real-world orientation of the probe mounted to the manipulator. This ensures that the clinician's movements remain intuitive relative to the real-time ultrasound image orientation. This is important for clinical transparency and cannot be addressed by simpler, linear-only force feedback devices.

When the robotic ultrasound system is guided by an operator through a haptic device, ensuring safety and stability in the interaction is of the utmost importance. Haptic–robot interaction is sensitive to delays, especially when communicating over remote connections. To mitigate these effects while maintaining high-fidelity six-DoF interaction, we implement the dual-proxy paradigm for remote operations (Jorda et al., 2022). This method was validated over a distance of 10,000 km to demonstrate its stability and robustness to communication delays.

### 4.1. Haptic–robot interaction via the dual-proxy model

The dual-proxy model, illustrated in Figure 7, captures the behavior of the robot and haptic device and relates the two

workspaces through an affine transformation along with a constant scaling factor. The haptic device position, $x_H$, and rotation, $R_H$, are measured in the device frame. These values are expressed in the robot frame to obtain the robot proxy position and rotation.

$$x_{rp} = x_{R-CW} + sR_{H-R}^T x_H, \tag{26}$$

$$R_{rp} = R_{H-R}^T R_H R_{H-R} R_{R-CW}, \tag{27}$$

where $x_{R-CW}$ and $R_{R-CW}$ are the offsets between the robot and the desired center of the robot workspace in position and rotation, respectively, $s$ is the scaling factor, and $R_{H-R}$ is the rotation matrix from the robot frame to the haptic device frame. The reverse transformation computes the haptic proxy position and rotation

$$x_{hp} = R_{H-R}(x_R - x_{R-CW})s^{-1} \tag{28}$$

$$R_{hp} = R_{H-R} R_R R_{R-CW}^T R_{H-R}^T. \tag{29}$$

from the robot position $x_R$ and rotation $R_R$ measured in the robot's base frame.

The control inputs are computed by projecting the robot and haptic proxies onto the task motion and force spaces, which are defined by the robot's directions of free motion and constraints. The robot's task specification matrices are described in Section 3.8.2. After computing the robot's force projection matrix, $\Omega_f$, the haptic device force-space projection matrix is

$$\Omega_{f,H} = \begin{bmatrix} R_{H-R} & 0_{3\times3} \\ 0_{3\times3} & R_{H-R} \end{bmatrix} \Omega_f \begin{bmatrix} R_{H-R}^T & 0_{3\times3} \\ 0_{3\times3} & R_{H-R}^T \end{bmatrix}, \tag{30}$$

and the haptic device motion-space projection matrix is $\Omega_{m,H} = I_{6\times6} - \Omega_{f,H}$.

In motion space, the robot and haptic proxies represent goal points. In force space, a target force is computed with the position error between the robot/haptic current position and robot/haptic proxy using virtual springs with stiffness. The control inputs to the robot controllers (15) and (19) are

$$x_{d,t} = x_{rp}, \tag{31}$$

$$R_d = R_{rp}, \tag{32}$$

$$F_d = -K_{vir,R} \begin{bmatrix} x_R - x_{rp} \\ \delta\phi_R \end{bmatrix}. \tag{33}$$

where $K_{vir,R}$ encodes the robot's virtual spring stiffnesses and $\delta\phi_R$ is the instantaneous angular error between the end-effector's actual and desired orientations.

The haptic control input, with $K_{vir,H}$ as the haptic virtual spring stiffness and $\delta\phi_H$ as the instantaneous angular error between the actual and desired haptic device orientation, is

$$F_H = -K_{vir,H} \begin{bmatrix} x_H - x_{hp} \\ \delta\phi_H \end{bmatrix}. \tag{34}$$

On a haptic device capable of rendering feedback moments, the operator feels a torsional spring restoring the haptic device to the same orientation as the actual transducer mounted on the robot. This restoring moment is depicted in Figure 8.

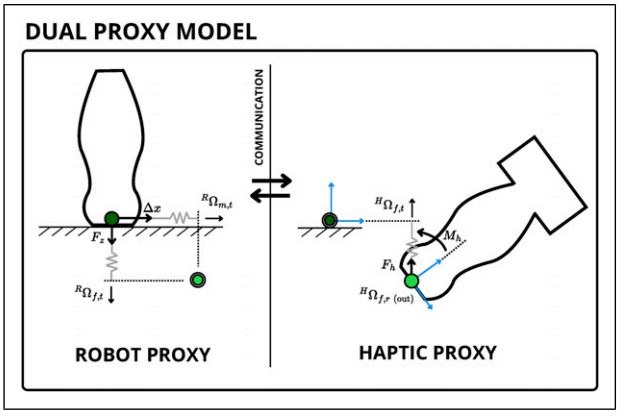



**Figure 7. Left**: Interaction between robot transducer and robot proxy in the force/motion spaces. **Right**: Interaction between haptic device and haptic proxy in the force/motion spaces.

### 4.2. Online environmental contact estimation

In an ultrasound setting, the haptic feedback fidelity relies on knowledge of the robot environment geometry, which is estimated through the force-space particle filter (Jorda et al., 2022). In free space, most of the particles are grouped in the center of a 3D unit sphere as shown in Figure 9. However, when the end-effector is in contact, most particles are along the sphere boundary and grouped around the normal to said contact. This binary classification between free space and contact is used to parameterize the robot controller as specified in Section 3.8.2.

## 5. Experimental validation

We characterize the performance of the proposed explicit force control (EFC) framework by first evaluating its contact force regulation against a state-of-the-art implicit method based on integral adaptive admittance control (IAAC) (Jiang et al., 2024).

Then, we demonstrate the system's broader utility through simultaneous contact force and force-based orientation optimization across several clinically-inspired scenarios. These evaluations include static imaging of hard and soft tissue to demonstrate force and moment control engagement in different regions, and dynamic scanning tasks that emulate clinical sweep motions.

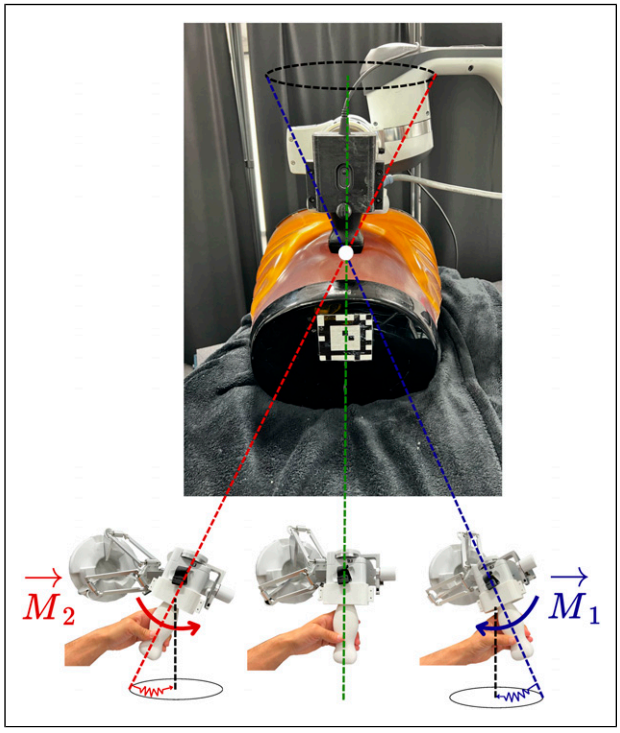



**Figure 8.** Haptic moment feedback.

Afterward, we investigate the framework's motion-aware and force-aware capabilities by simulating patient movement during a scan. Specifically, these two experiments test the controller's ability to track perturbations in motion directions while simultaneously compensating for environmental shifts in the explicit force- and moment-controlled directions.

Finally, we assess the system in a teleoperation context by comparing a manual six-DoF haptic scan, in which the operator specifies varying axial forces, to an autonomous replay of the same recorded scan. In the replay task, the EFC autonomously regulates the axial force to a constant 13.4 N, and the in-plane and out-of-plane moments to zero, while tracking the recorded trajectory that is projected into the motion-controlled directions. This comparison between the two haptically-guided scanning modes (manual and autonomous replay) allows us to quantify the differences in acquired image confidence.

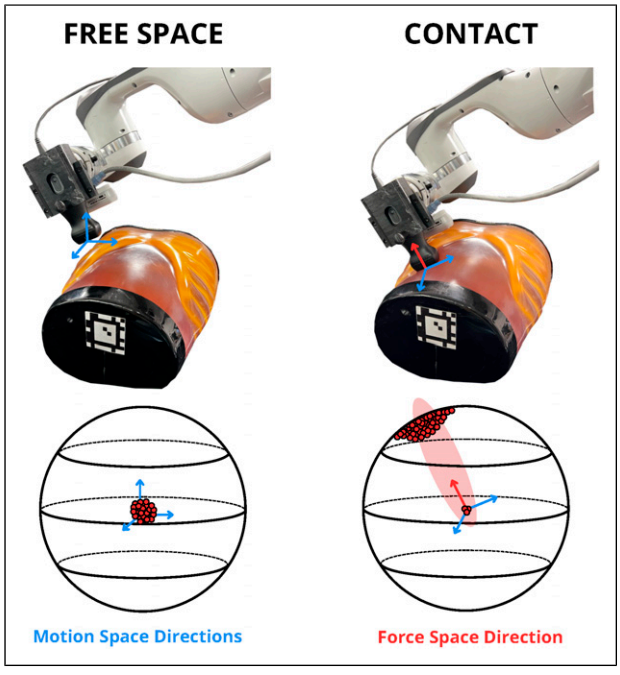



**Figure 9.** Online environmental contact estimation with the force-space particle filter.

## 5.1. Experimental setup

All experiments were conducted using the torque-controlled seven-DoF Franka Emika Panda manipulator (Franka Robotics GmbH) and the high-fidelity ABDFAN abdominal phantom (Kyoto Kagaku Co., Ltd.), as illustrated in Figure 10. The phantom provides clinically realistic echogenicity and includes internal structures such as the liver, gallbladder, kidneys, spleen, vascular structures, and pathologies such as cysts and tumors.

To facilitate repeatable and clinically relevant trajectories, we define five anatomical landmarks (A to E) across the torso surface. Points A and D are positioned over the rigid rib structures to evaluate hard-tissue interaction, while B and C are located over compliant soft-tissue regions. Point E serves as the central reference, which is also above a soft-tissue region. As shown in the schematic on the right of Figure 10, these landmarks are organized into two distinct layers: a contact plane that corresponds to the phantom's coronal plane, and a free-space plane located 10 cm above.

During all experiments, we define task-space coordinates resolved in the transducer frame: axial force represents contact force along the transducer's long axis, $z$, while in-plane and out-of-plane moments represent torques measured about the transducer's $x$ and $y$ axes, respectively. Unless otherwise indicated, the controller is set to maintain a constant axial force of 13.4 N and zero-magnitude moments. Robotic and haptic data were recorded at 100 Hz, while ultrasound images were recorded at 5 Hz.

*5.1.1. Robot end-effector design.* The end-effector design for robotic ultrasound scan acquisition, based on the one from our previous work (Piedra et al., 2024), is presented in Figure 11. The force, in-plane, and out-of-plane axes in the transducer frame are labeled. The force/torque sensor is used to (1) maintain constant contact between the ultrasound transducer and phantom surface and (2) align the transducer

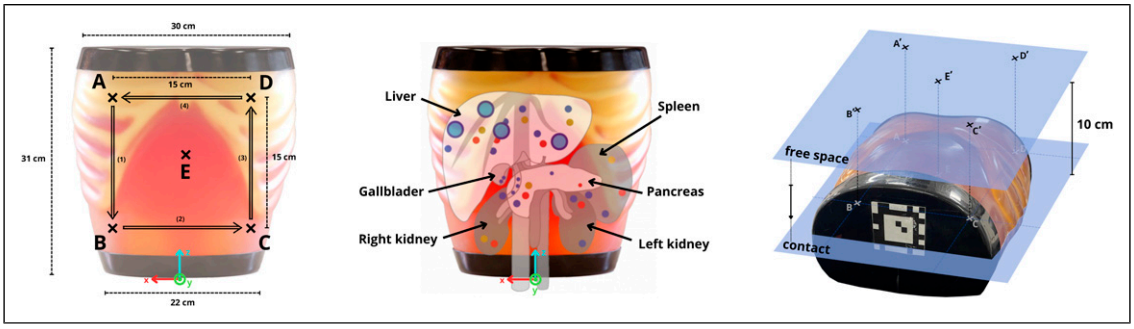



**Figure 10. Left**: Dimensions of the phantom and experimental paths. **Middle**: Anatomy of the torso. **Right**: Presentation of the free space and contact planes containing the via points and their projections inside the phantom.

lens normal to the surface through zero-moment control. The specific force/torque sensor used in the experiments is a 9105-NET-GAMMA1715 (ATI Industrial Automation) with calibration SI-130-10, which has a force resolution of $\frac{1}{20}$N along the transducer's force axis and a moment resolution of $\frac{1}{800}$N·m.

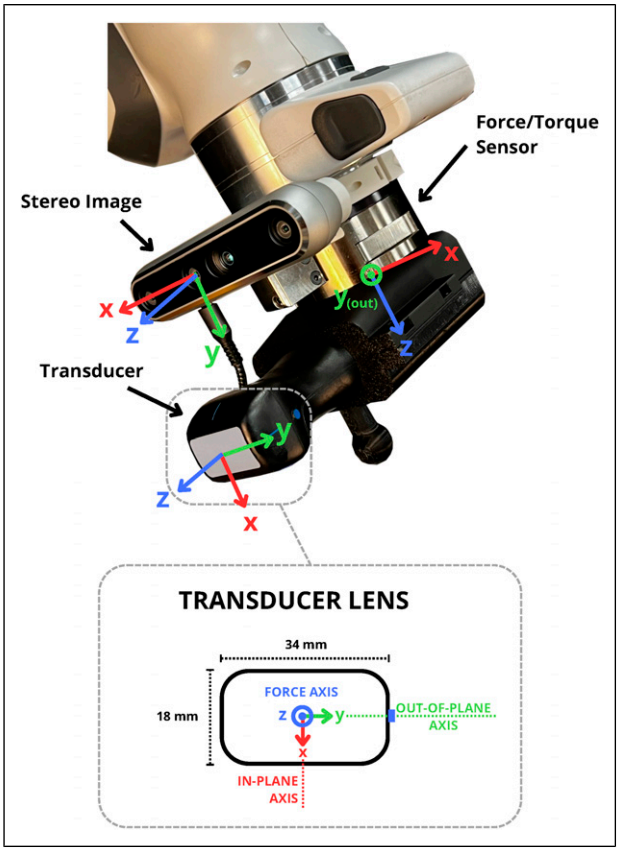



**Figure 11. Top**: Robot end-effector design including force/torque sensor, stereo camera, and ultrasound transducer. **Bottom**: Transducer frame with labeled axes and lens dimensions.

*5.1.2. Tracking phantom motion.* To ensure continuous tracking of the phantom's position and orientation, we utilize a world camera that monitors the phantom's RAS anatomical coordinate frame. This global tracking is necessary because the robot's end-effector camera (shown in Figure 11), while useful for registration and other applications, is subject to occlusions during the scanning process. Both the world camera and the end-effector camera are Intel RealSense D455 RGB-D cameras.

To utilize the phantom tracking information for robotic control, the world camera's pose must be registered with respect to the robot base frame. This is achieved by simultaneously capturing a fixed AprilTag (Olson, 2011) on the phantom with both the world camera and the end-effector camera. Since the end-effector camera and the ultrasound transducer are both rigidly fixed to the robot, their poses relative to the robot base are known via forward kinematics. Consequently, the spatial transform between the world camera and the robot base is computed through the shared AprilTag reference, as illustrated in Figure 12.

*5.1.3 Haptic interface and control definitions.* For haptically-guided scans, we utilize a Force Dimension sigma.7 haptic device shown in Figure 2, which provides six-DoF force/torque feedback (Tobergte et al., 2011). This interface allows the operator to perceive the physical interaction forces generated during the robotic ultrasound procedure. To ensure clinical realism, a custom 3D-printed adapter was developed to mimic the ergonomic form factor of a standard ultrasound transducer, as shown in Figure 13. The control framework maps the motion of the haptic stylus directly to the robot's desired motion and/or force, which is then regulated by our controller.

The modes of operation, illustrated in Figure 14, are categorized based on the task inputs specified by the human user or the robot. In a fully manual scan, the user specifies the desired motion, force, and orientation. In a fully autonomous scan, the robot specifies all of these

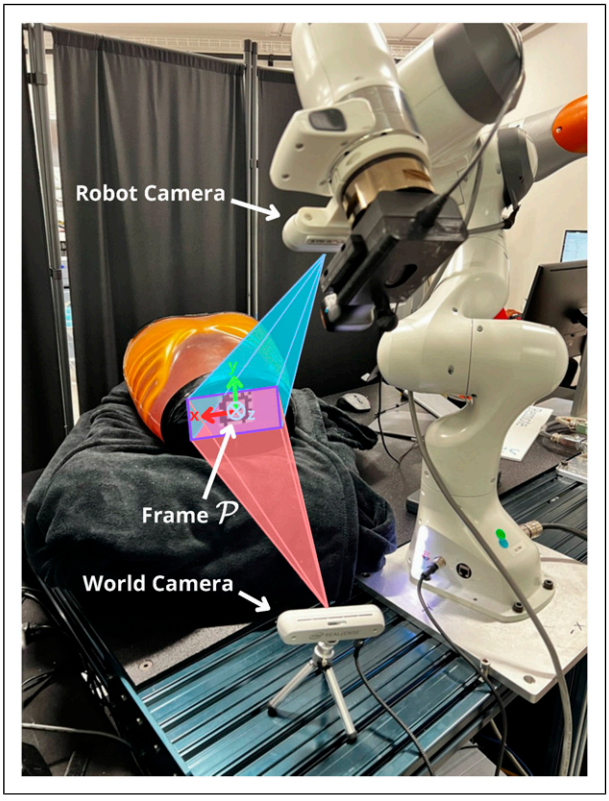



**Figure 12.** Robot end-effector and world cameras simultaneously tracking phantom anatomical frame.

parameters instead (with the orientation partially determined through explicit moment control). A semi-autonomous scan combines both approaches, where the user specifies the desired translational motion and the desired imaging plane, while the robot specifies the desired force and controls the remaining orientation through explicit moment control.

For the purpose of comparison a manual haptically-guided scan to an assisted semi-autonomous one, we implemented two primary operational modes:

- **Manual haptic scan**: The user directly controls the robot proxy through the haptic interface. In this mode, our controller regulates the axial contact force based on the user's input, but moment control is disabled. This allows the operator to freely manipulate the transducer orientation.
- **Semi-autonomous replay**: The robot autonomously follows a previously recorded haptic scanning path. In this mode, both explicit force and moment control are active. The robot maintains a constant target force of 13.4 N and utilizes moment control to maintain the transducer perpendicularity relative to the surface normal, regardless of the original recorded orientation.

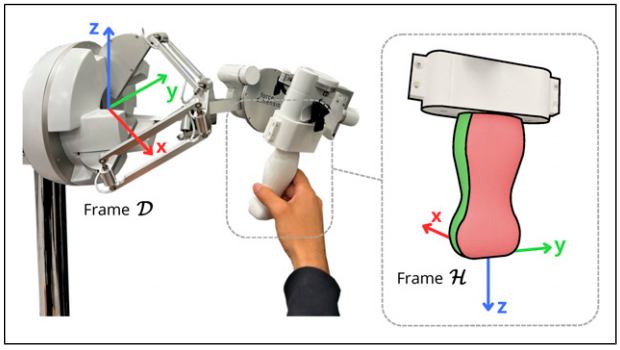



**Figure 13. Left**: Haptic device frame attached to the base of the six-DoF haptic device. **Right**: Haptic probe adapter frame.

*5.1.4. Assessing image quality with confidence maps.* The convex ultrasound images were captured with a Butterfly iQ+ transducer, which operates through Butterfly iQ's proprietary software. The imaging software provides access to 2D brightness (B-mode) images, but does not provide the raw radio frequency signals along the scanlines.

Following the methodology described in Karamalis et al. (2012), we compute the confidence of each acquired image through a multi-step pipeline shown in Figure 15. The procedure involves the following steps: (1) record the Butterfly iQ screen image; (2) mask the pixels in the screen image corresponding to the convex ultrasound image; (3) flatten the convex masked pixels to a rectangular shape; (4) compute the confidence map for the flattened image; (5) for illustration purposes, warp the flattened confidence map back to the original convex shape of the ultrasound image.

The algorithm utilizes the free parameters in Karamalis et al. (2012): $\alpha = 6$, $\beta = 90$, and $\gamma = 0.05$. The $\alpha$ parameter scales intensities to model depth-dependent signal attenuation, $\beta$ regulates the sensitivity of the graph weights to image features representing acoustic transmission, and $\gamma$ penalizes lateral discontinuities to account for the physical constraints of ultrasound beam-width. These confidence maps provide a quantitative metric of image quality based on acoustic coupling, which is optimized when the probe is flush with the tissue surface.

## 5.2. Benchmarking explicit force control

This section presents a quantitative assessment of the axial force regulation performance of the proposed EFC architecture. To establish a baseline for comparison, the EFC is benchmarked against the IAAC scheme, representing a state-of-the-art implicit force control strategy. The primary objective of this benchmarking study is to characterize the force tracking accuracy of both controllers under dynamic interaction conditions. The evaluation is conducted across two distinct experimental scenarios:

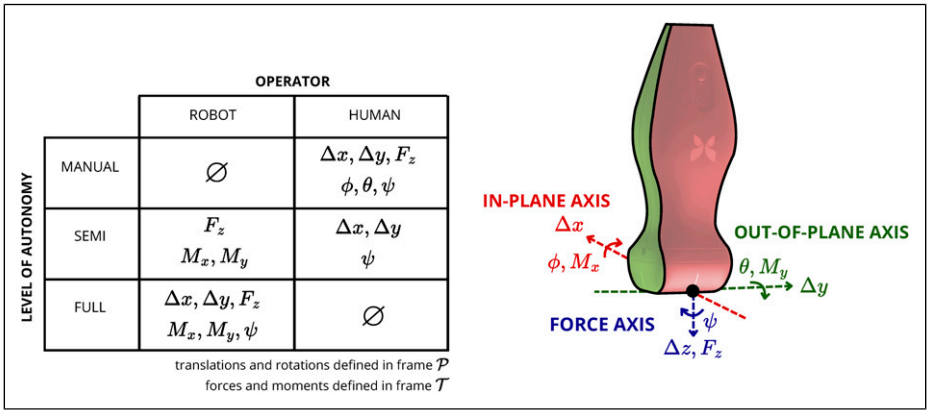



**Figure 14. Left**: Modes of haptically-guided operations. **Right**: Definition of transducer frame directions.

- **Sinusoidal force tracking**: Assessing the controller's ability to follow a time-varying force profile, which is critical for applications requiring precise force modulation.
- **Constant force tracking under phantom motion**: Maintaining a constant target force while the phantom is subjected to external translational motion, simulating a moving environment such as patient respiration or shifting.

*5.2.1. Explicit force controller design.* The explicit force controller is implemented following the details presented in Section 3. The gains used for our explicit motion and force controller are listed in Table 1. The velocity gain matrix presented in equation (16) is given by the following values:

$$K_{v,f} = \begin{bmatrix} K_{v,f,t} I_{3x3} & 0_{3x3} \\ 0_{3x3} & K_{v,f,r} I_{3x3} \end{bmatrix} \tag{35}$$

where $K_{v,f,t}$ and $K_{v,f,r}$ are the translational and rotational force-space velocity gains, respectively.

Similarly, the translational and rotational motion gain matrices for proportional and derivative feedback in equation (19) are given by:

$$K_{p,m} = \begin{bmatrix} K_{p,m,t} I_{3x3} & 0_{3x3} \\ 0_{3x3} & K_{p,m,r} I_{3x3} \end{bmatrix}, \tag{36}$$

$$K_{v,m} = \begin{bmatrix} K_{v,m,t} I_{3x3} & 0_{3x3} \\ 0_{3x3} & K_{v,m,r} I_{3x3} \end{bmatrix}. \tag{37}$$

Note that the compliant experimental environment allows for large proportional force and moment control gains. The small transducer lens moment arm further permits a large proportional moment control gains. Additionally, we implement zero-crossing reset for the integral terms in both force and moment control to prevent integral windup.

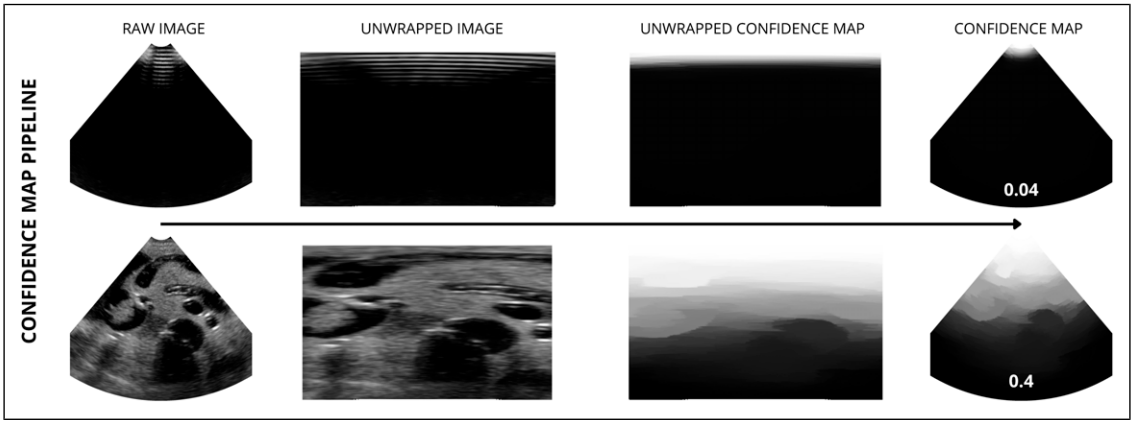



**Figure 15.** Confidence map generation pipeline. **Left**: Raw ultrasound images acquired using the Butterfly iQ software. **Middle left**: Corresponding flattened ultrasound image. **Middle right**: Confidence map computed from the flattened image. **Right**: Final convex (warped) confidence map.

*5.2.2. Implicit force controller design.* We implemented a state-of-the-art implicit IAAC contact force controller as a baseline for comparison to our EFC approach. The details of the IAAC implementation are described in Jiang et al. (2024), and the controller is summarized in Algorithm 1. Two versions of the IAAC controller are presented: one with the baseline parameters used in Jiang et al. (2024) for robotic ultrasound on a phantom, and another with parameters tuned for improved performance on our experimental setup. The baseline parameters are as follows: $m = 4$ kg, $b = 480$ N·s·m$^{-1}$, $k = 200$ N·m$^{-1}$, $\widehat{k}_e = 1000$ N·m$^{-1}$, $\alpha = 1$, $\beta = 1$, $k_i = 0.2$. Tuning the response of the IAAC for a sinusoidal force goal resulted in the following modified parameters: $b^* = 120$ N·s·m$^{-1}$, $k^* = 100$ N·m$^{-1}$, $\widehat{k}_e^* = 2000$ N·m$^{-1}$, $k_i^* = 0.8$ .

**Algorithm 1** Integral adaptive admittance control.

**Input:** Sensed force $F_s$, desired force $F_d$, end-effector position $x$, mass $m$, stiffness gain $k$, damping gain $b$, integral gain $k_i$, initial environmental contact position estimate $\hat{x}_e$, initial environmental stiffness estimate $\hat{k}_e$, adaptive rates $\alpha$ and $\beta$, time step $\Delta t$

**Output:** Desired robot end-effector position $x_d$

$x_{ref} \leftarrow x$ ▷ *Record initial contact position*

**while** in contact **do**

  $F_e \leftarrow F_s - F_d$ ▷ *Compute force error*

  $\hat{F}_e \leftarrow \hat{k}_e(\hat{x}_e - x)$ ▷ *Estimate force error*

  $\hat{k}_e \leftarrow \hat{k}_e + \alpha \cdot x \cdot (\hat{F}_e - F_e) \cdot \Delta t$ ▷ *Update estimated environmental stiffness*

  $\hat{x}_e \leftarrow \hat{x}_e - \beta \cdot \frac{\hat{F}_e - F_e}{\hat{k}_e} \cdot (\frac{\alpha}{\beta} \cdot x \cdot \hat{x}_e + 1) \cdot \Delta t$ ▷ *Update estimated environmental contact position*

  $F_i \leftarrow F_i + F_e \cdot \Delta t$ ▷ *Update integral term*

  $\Delta\ddot{x} \leftarrow \frac{F_e + k_i F_i - b\Delta\dot{x} - k\Delta x}{m}$ ▷ *Substitute terms into admittance model*

  $\Delta\dot{x} \leftarrow \Delta\dot{x} + (\Delta\ddot{x} \cdot \Delta t)$

  $\Delta x \leftarrow \Delta x + (\Delta\dot{x} \cdot \Delta t)$

  $x_d \leftarrow x_{ref} + \Delta x$ ▷ *Update desired position*

  **return** $x_d$

*5.2.3. Experimental design.* In order to compare the contact force control performance of our EFC against an IAAC, the robot followed sinusoidal force trajectories from 0 N to 20 N, covering a typical range of forces in clinical ultrasound of the abdomen. The force trajectories had increasing frequencies of 0.2 Hz, 0.4 Hz, 0.8 Hz, and 1.6 Hz, exceeding the typical human respiratory frequency of around 0.3 Hz. Contact force control was maintained during experiments at anatomical landmark E on the torso phantom.

A second set of experiments for comparing the contact force controllers involved following a constant desired force of 13.4 N while the phantom was repeatedly displaced toward the transducer and away from the transducer by a distance of 15 cm with a constant velocity of about 5 cm·s$^{-1}$.

**Table 1.** Controller gains for explicit force and motion control.

| Force control | | | Motion control | | |
|---|---|---|---|---|---|
| Parameter | Value | Unit | Parameter | Value | Unit |
| $K_{p,f,t}$ | 8 | — | $K_{p,m,t}$ | 250 | N·m$^{-1}$ |
| $K_{v,f,t}$ | 25 | N·s·m$^{-1}$ | $K_{v,m,t}$ | 24 | N·s·m$^{-1}$ |
| $K_{i,f,t}$ | 5 | s$^{-1}$ | | | |
| $K_{p,f,r}$ | 25 | — | $K_{p,m,r}$ | 200 | N·rad$^{-1}$ |
| $K_{v,f,r}$ | 25 | N·s$^{-1}$ | $K_{v,m,r}$ | 23 | N·s·rad$^{-1}$ |
| $K_{i,f,r}$ | 5 | s$^{-1}$ | | | |

The linear actuator used for this set of experiments exhibited asymmetric accelerations in the two different directions, resulting in slightly different time durations for the approach and retraction motions. The phantom stayed at rest for approximately 5 s between the periods of motion.

*5.2.4. Results on sinusoidal force tracking.* The results for the dynamic sinusoidal force tracking are illustrated in Figure 16 and detailed in Table 2. Across the tested frequency spectrum (0.2 Hz to 1.6 Hz), the EFC maintained a significantly lower phase lag compared to the IAAC variants. Specifically, the EFC exhibited a lag ranging from 1.44° at 0.2 Hz to a maximum of only 5.76° at 1.6 Hz. In contrast, the tuned IAAC and baseline IAAC reached substantial phase lags of 121.0° and 149.8°, respectively, at the highest frequency.

In terms of tracking accuracy, the EFC achieved sub-Newton average errors for frequencies up to 0.8 Hz (0.56 ± 0.28 N), with a maximum recorded error of 1.28 N. At 1.6 Hz, the EFC recorded an average error of 1.01 ± 0.61 N and a peak error of 3.37 N. This performance represents a significant improvement over the tuned IAAC, whose average error rose from 1.32 ± 0.72 N at 0.2 Hz to 11.28 ± 5.39 N at 1.6 Hz.

*5.2.5. Results on constant force tracking under phantom motion.* The tracking performance during repetitive phantom displacement is visualized in Figure 17 and summarized in Table 3. During the time periods in which the phantom was stationary, both the EFC and the tuned IAAC performed effectively, yielding average errors of 0.19 ± 0.22 N and 0.29 ± 0.73 N, respectively.

However, the EFC demonstrated superior consistency during active environmental motion. It maintained an average tracking error of 1.25 ± 0.37 N (max 1.66 N) when the phantom moved toward the transducer and 1.43 ± 0.45 N (max 2.02 N) during retraction. Under the same dynamic conditions, the tuned IAAC average error increased to approximately 5.4 N across both directions, while the baseline IAAC reached a peak error of 18.76 N during the approach phase.

## 5.3. Static force and moment control

The objective of this experiment is to evaluate the system's ability to maintain a stable acoustic window and

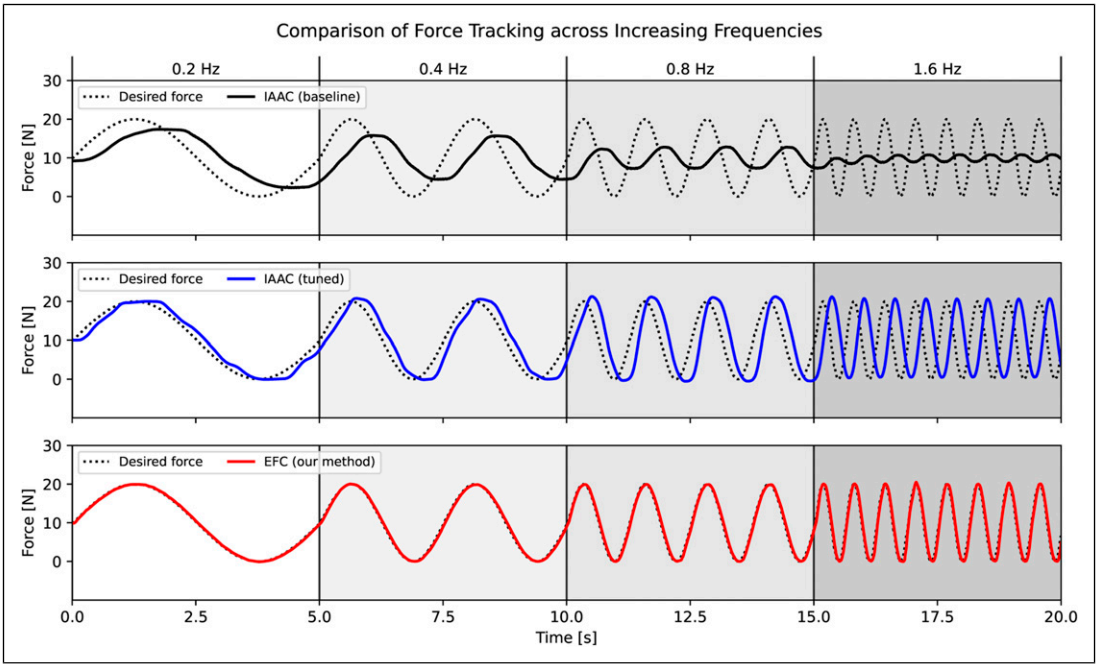



**Figure 16.** Sensed contact force during the sinusoidal force tracking experiment, represented in the transducer frame. The sinusoidal force frequency increases from 0.2 Hz to 1.6 Hz (left-to-right).

demonstrate how each component of the EFC framework contributes to image quality. In clinical practice, sonographers often hold the transducer stationary to observe blood flow or organ motility; however, maintaining a constant force and perpendicular orientation on curved, compliant anatomy is manually taxing. This section demonstrates the EFC's ability to regulate a constant 13.4 N axial force while simultaneously utilizing moment control to ensure the transducer remains flush with the phantom surface at landmarks A–E.

*5.3.1. Experimental design.* The static experiment was divided into five sequential phases and repeated for each of the via points A, B, C, and D. Figure 10 highlights the positions of these points, along with their projections onto a plane located above the phantom surface. This projection allowed the robot to align vertically above each target point while remaining in free space. In Phase I, the robot moved to point E′ in free space. In Phase II, the robot translated to the top projection point A′, B′, C′, or D′. In Phase III, the robot tried to reach the points A, B, C, or D located within the phantom and made contact with the phantom surface using position control. In Phase IV, force control was engaged to apply a target force of 13.4 N. In Phase V, moment control was activated in addition to force control to align the probe with the surface normal. This process was repeated for each target point and ultrasound images were recorded for computing corresponding confidence maps, as described in Section 5.1.4.

*5.3.2. Results.* Figure 18 presents the sensed forces and moments expressed in the transducer frame, along with sample images in the lower right showing the probe before and after moment control is applied. In the initial two phases, forces and moments remain near zero, consistent with free-space motion. Correspondingly, the average confidence is also near zero, reflecting poor image quality when the probe is not in contact. In Phase III, pose control

**Table 2.** Benchmarking contact force controllers with respect to phase lag and force tracking error across different frequencies.

| | Phase lag [°] | | | Force tracking error (avg. ± std. dev./max) [N] | | |
|---|---|---|---|---|---|---|
| Freq (Hz) | IAAC (baseline) | IAAC (tuned) | EFC (our method) | IAAC (baseline) | IAAC (tuned) | EFC (our method) |
| 0.2 | 33.8 | 11.52 | 1.44 | 3.77 ± 1.79/6.21 | 1.32 ± 0.72/2.40 | 0.19 ± 0.10/0.44 |
| 0.4 | 64.8 | 23.0 | 2.88 | 6.14 ± 3.03/10.28 | 2.78 ± 1.35/5.44 | 0.32 ± 0.19/0.83 |
| 0.8 | 109.4 | 49.0 | 5.76 | 7.31 ± 3.59/12.97 | 5.65 ± 3.07/11.30 | 0.56 ± 0.28/1.28 |
| 1.6 | 149.8 | 121.0 | 5.76 | 6.96 ± 3.38/12.51 | 11.28 ± 5.39/18.13 | 1.01 ± 0.61/3.37 |

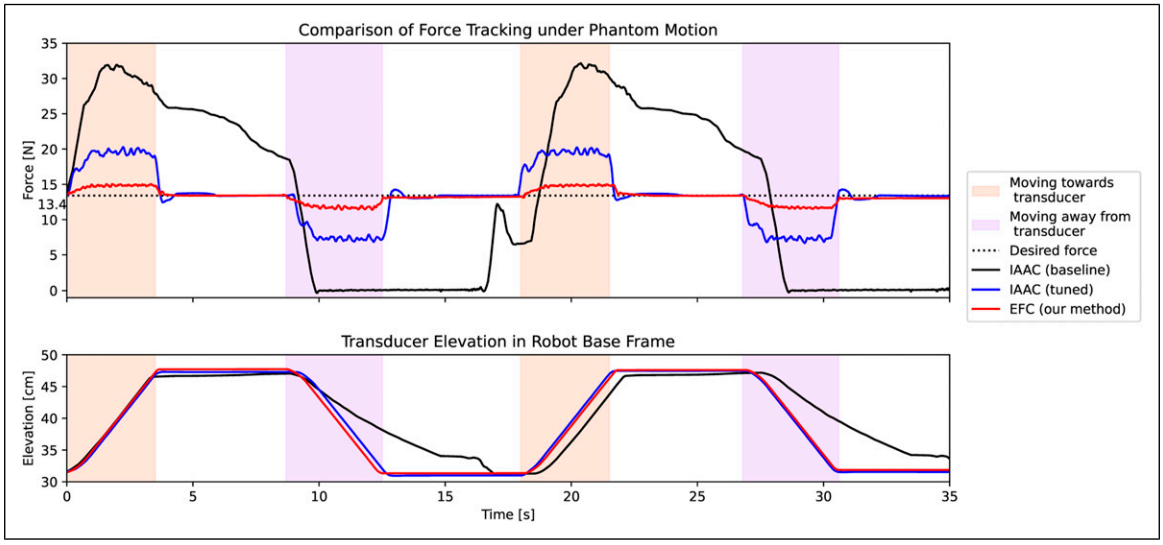



**Figure 17. Top**: Sensed contact force, represented in the transducer frame, during the constant force tracking experiment under phantom motion. The shaded regions indicate the time periods during which the phantom was moving toward or away from the transducer. The baseline IAAC occasionally loses contact with the phantom during motion, resulting in periods of zero sensed force. **Bottom**: The transducer's elevation in the robot base frame. The baseline IAAC occasionally loses contact with the phantom and therefore does not follow the same elevation trajectory.

results in a sharp increase in both force and moment magnitudes, as contact occurs without regulation. Confidence remains low, except at point B, where partial alignment may have yielded a slight improvement. In Phase IV, force control is activated, leading to convergence toward the target contact force of 13.4 N, while moments remain nonzero. This yields a modest increase in image confidence. In Phase V, moment control is engaged, and the moments converge to zero. Simultaneously, image confidence increases significantly, indicating improved probe alignment and image quality. The desired force remains stable throughout. At the bottom of the figure, raw ultrasound images and corresponding confidence maps illustrate representative examples from each phase.

Figure 19 shows the 3D trajectory of the end-effector (large dots) and the position proxy (small dots), both expressed in the phantom frame. The small dots represent the desired position, which attempts to bring the probe inside the phantom, while the end-effector remains constrained to the surface. The right plot shows the trajectory projected onto the phantom surface in the background, with the color gradient from blue to red indicating progression over time.

The results for this and the subsequent experiments are summarized in Table 4.

## 5.4. Dynamic force and moment control

To capture volumetric data or perform longitudinal scans, the robot must maintain contact parameters while traversing unknown or varying surface curvatures. This experiment evaluates the tracking performance during a continuous sweep across the abdominal surface of the phantom. The primary challenge addressed here is the simultaneous regulation of axial force and the suppression of in-plane and out-of-plane moments as the tissue stiffness and surface normal change.

*5.4.1. Experimental design.* In the dynamic experiment, the robot followed the closed-loop path depicted in the left image of Figure 10, starting and ending at point A. This

**Table 3.** Benchmarking contact force controllers with respect to force tracking error under different types of phantom motion.

| | Force tracking error (avg. ± std. dev./max) [N] | | |
|---|---|---|---|
| Controller | Stationary | Toward | Away |
| IAAC (baseline) | 11.54 ± 2.83/16.74 | 13.59 ± 5.46/18.76 | 10.46 ± 4.25/13.73 |
| IAAC (tuned) | 0.29 ± 0.73/6.18 | 5.61 ± 1.21/6.88 | 5.22 ± 1.71/6.73 |
| EFC (our method) | 0.19 ± 0.22/1.52 | 1.25 ± 0.37/1.66 | 1.43 ± 0.45/2.02 |

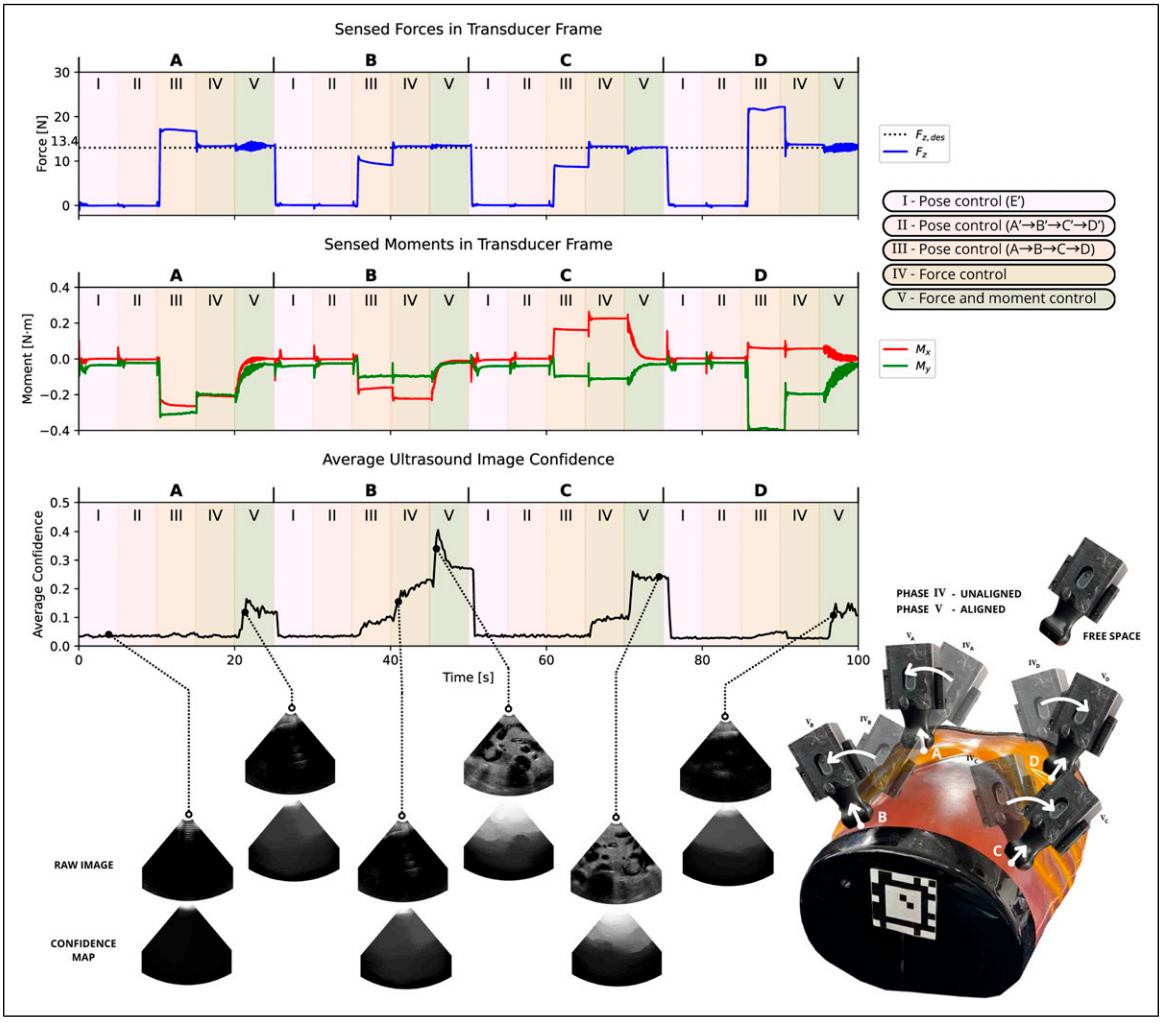



**Figure 18. Top**: Sensed forces, moments (in the transducer frame), and average image confidence over time across all experimental phases (described in legend). **Bottom left**: Representative raw ultrasound images and corresponding confidence maps at selected points. **Bottom right**: Visual depiction of the final two phases at points A, B, C, and D, showing the probe in contact with the phantom.

trajectory was designed to guide the transducer across regions of varying stiffness, including both stiff and soft sections, and transitions in both directions. The commanded scanning speed was 3 cm·s$^{-1}$.

*5.4.2. Results.* The force plot in Figure 20 shows that the sensed force in the transducer frame along the controlled direction remains close to the target value of 13.4 N, with an average error of 0.33 N. The average moments about the $x$ and $y$ axes in the transducer frame are $-0.01$ N·m and 0.02 N·m, respectively. Notably, transitions between soft and stiff regions of the phantom do not significantly affect the stability of the controlled force or moment regulation.

In Figure 21, the left plot displays the 3D trajectory of the end-effector in the phantom frame, along with the robot proxy it follows. The average position tracking error is 0.75 cm. The right panel confirms that the trajectory closely follows the intended square path marked on the phantom surface, with slight erroneous curve tracking due to the phantom's curved surface. The average orientation error between the actual and desired transducer orientation is 1.27°.

## 5.5. Force-aware and motion-aware control

Clinical environments are inherently dynamic, with disturbances arising from patient respiration and shifting positions. This section explores the robustness of the EFC framework when subjected to external environmental motion. We evaluate the system's ability to decouple the desired interaction force from the translational disturbance of the phantom, ensuring that image quality and contact safety are preserved despite patient movement.

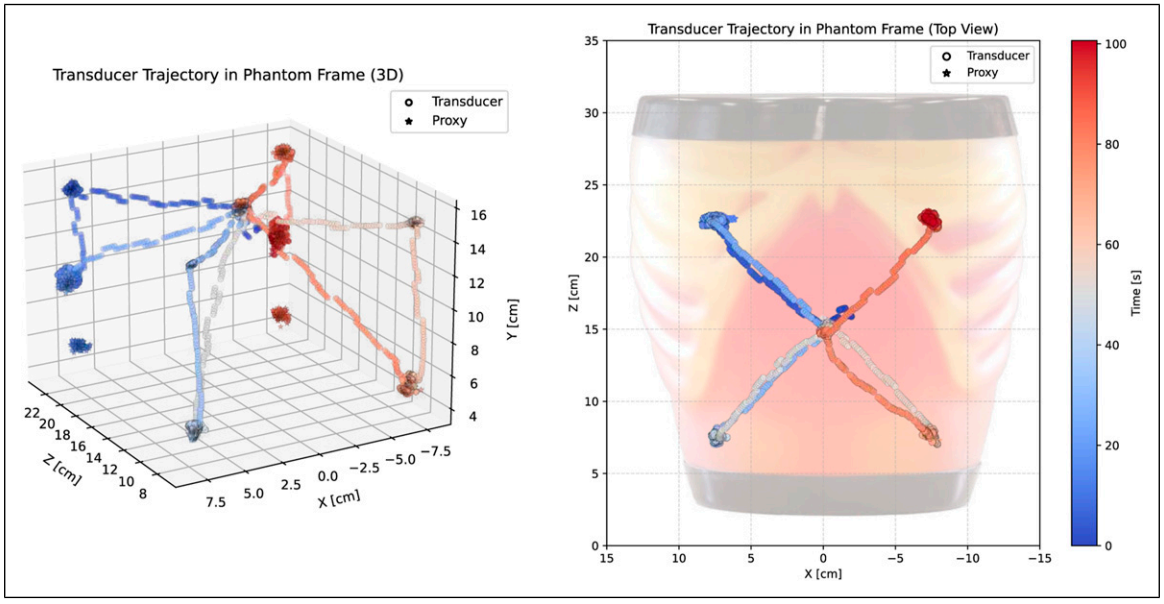



**Figure 19. Left**: 3D trajectories of the end-effector and robot proxy during the static experiment around the four target points. **Right**: 2D projection of the same trajectories onto the (*x*, z) plane, overlaid on a background image of the phantom.

*5.5.1. Experimental design.* Two experiments were designed to evaluate the robot's ability to maintain stable position, constant contact force, and zero moment despite patient motion in 3D space. In both experiments, the robot was commanded to reach point E inside the phantom using full force and moment control. In the motion-aware experiment, an external operator manually displaced the phantom within the horizontal plane, defined by the *x* and *z* axes of the RAS anatomical frame. In the force-aware experiment, the operator introduced perturbations along the vertical direction (*y*-axis in the RAS frame) as well as in both in-plane and out-of-plane orientations.

*5.5.2. Results.* During the motion-aware experiment, shown on the left side of Figure 22, the sensed forces in the transducer frame closely follow the desired constant force of 13.4 N, with an average error of 0.13 N. The corresponding sensed moments in the *x* and *y* directions remain near zero, with average values of −0.01 N·m and 0.01 N·m in the *x* and *y* directions, respectively, and standard deviations of 0.01 N·m in both directions.

In contrast, during the force-aware experiment (right side of Figure 22), the average force error in the *z* direction increases to 1.31 N, and the average moments stay around zero, but the standard deviations in the *x* and *y* directions rise

**Table 4.** Experimental results on explicit force and motion tracking across various scanning scenarios. Directions are specified in the transducer frame.

| Experiment | Avg. force error (z)[N] | Avg. moment (*x*, *y*) [N·m] | Std. dev. moment (*x*, *y*) [N·m] | Avg. position error [cm] | Avg. orientation error [◦] | Avg. image confidence |
|---|---|---|---|---|---|---|
| Force tracking | 0.70 | — | — | — | — | — |
| Dynamic | 0.33 | −0.01, 0.02 | — | 0.75 | 1.27 | — |
| Motion-aware | 0.13 | −0.01, 0.01 | — | 0.63 | 0.57 | — |
| Force-aware | 1.31 | 0.05, 0.02 | — | 0.44 | 0.78 | — |
| Manual haptic | 0.30 | −0.08, 0.05 | 0.16, 0.04 | 0.77 | 2.11 | 0.27 |
| Replay haptic | 0.18 | −0.01, 0.03 | 0.02, 0.03 | 0.45 | 2.13 | 0.32 |

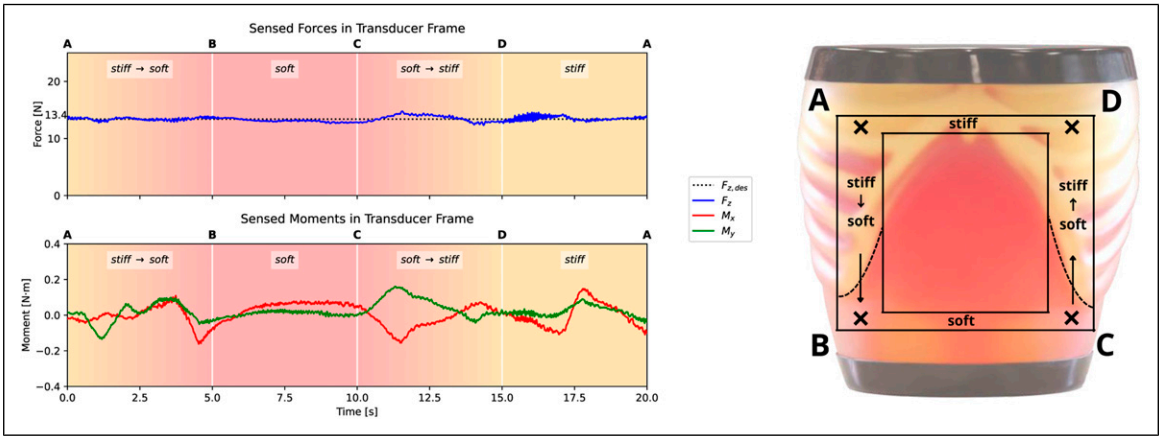



**Figure 20. Left**: Sensed forces and moments in the transducer frame recorded during the dynamic experiment. **Right**: Schematic of the robot's scanning path along the torso, indicating via points and relative surface stiffness.

to 0.05 N·m and 0.02 N·m, respectively. During the force-aware experiment, the phantom was subjected to both in-plane and out-of-plane displacements. This increase is attributed to the vertical motion of the phantom, as shown in the bottom photographs of the figure.

This vertical displacement is also evident in the 3D trajectories, which are plotted at the same scale in the $z$ direction for direct comparison. Additionally, the end-effector and robot proxy trajectories, shown in the third plot, indicate low tracking error in both experiments, with average positional errors of 0.63 cm and 0.44 cm and rotational errors of 0.57° and 0.78° for the motion-aware and force-aware experiments, respectively.

## 5.6. Haptically-guided scans

This final evaluation explores the synergy between human intuition and robotic precision through haptically-guided ultrasound scanning. We compare manual scanning (where the operator defines the transducer trajectory, force, and orientation) against a semi-autonomous replay mode that

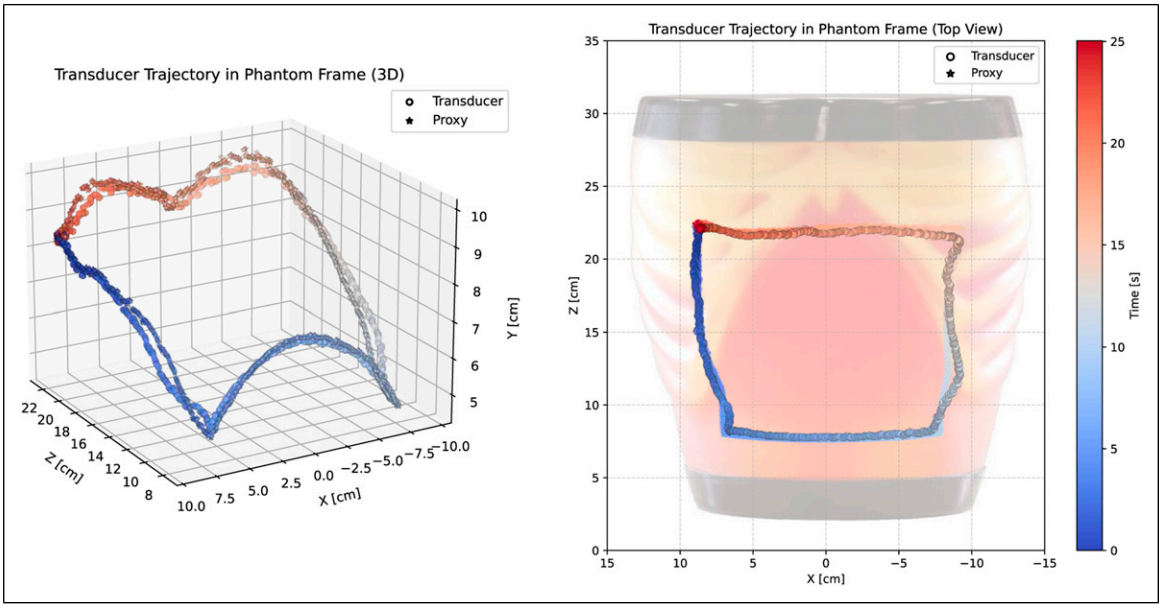



**Figure 21. Left**: 3D trajectories of the end-effector and robot proxy during the dynamic experiment around the four target points. **Right**: 2D projection of the same trajectories onto the ($x$, $z$) plane, overlaid on a background image of the phantom.

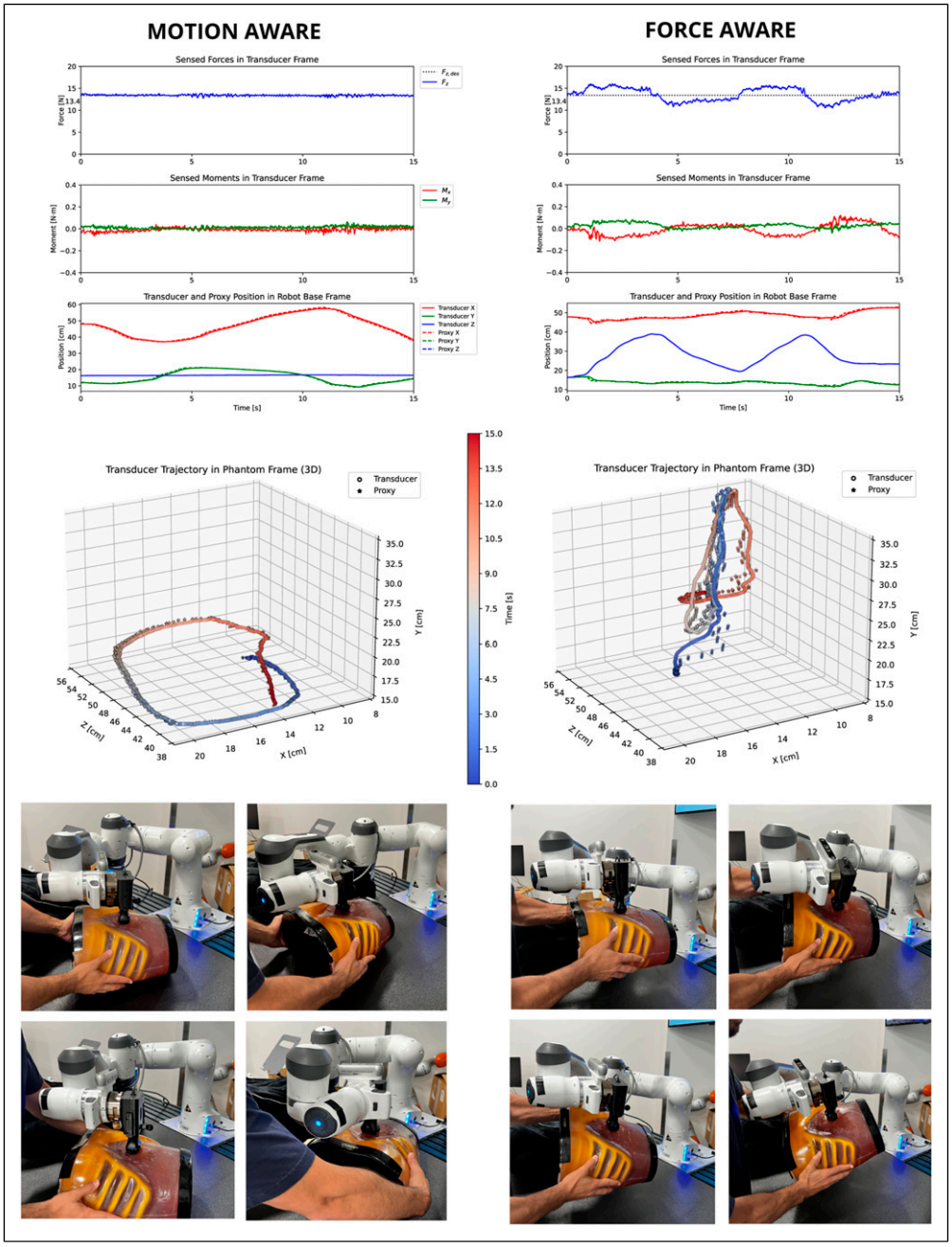



**Figure 22. Top**: Sensed forces and moments in the transducer frame, along with end-effector and proxy positions in the robot base frame for both motion-aware and force-aware experiments. **Middle**: 3D trajectories of the end-effector and proxy in the robot base frame during both experiments. **Bottom**: Photographic documentation illustrating the experimental setups for motion-aware and force-aware condition.

autonomously tracks a constant desired force and aligns the transducer normal to the surface.

*5.6.1. Experimental design.* This experiment involved performing a manual scan using the six-DoF haptic device shown in Figure 13. The robot followed a proxy controlled by the haptic device operator. Moment control was disabled during this phase, allowing the operator to freely adjust the probe's orientation. The proxy-following behavior was implemented as described in Section 4.1, using closed-loop force control.

The second part of the experiment consisted of replaying the same scan autonomously, with both force and moment control enabled. During this phase, the robot followed the recorded trajectory while actively regulating contact force and probe orientation using moment control to enhance

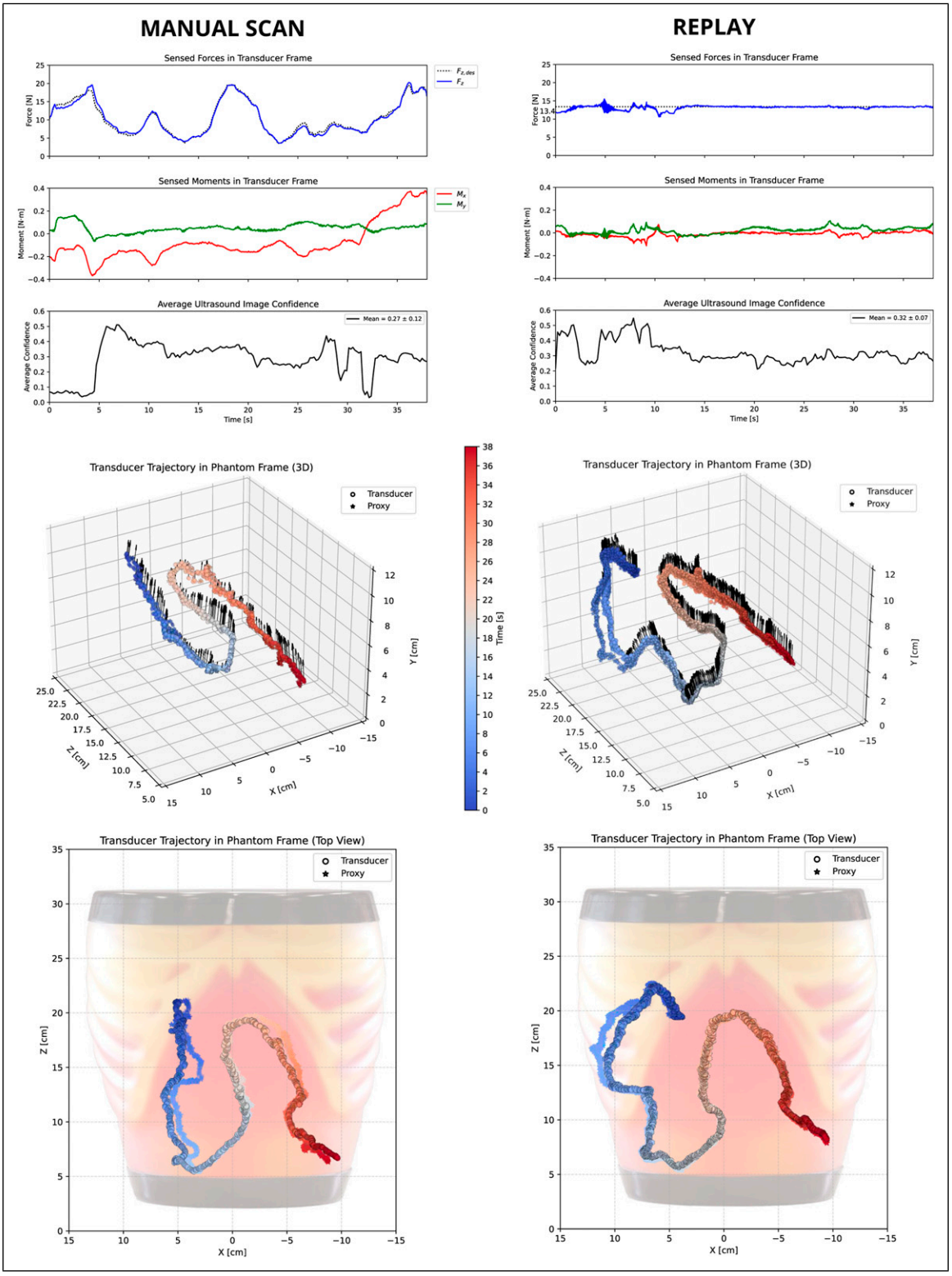



**Figure 23.** **Top**: Sensed forces, moments (in the transducer frame), and average image confidence over time for both manual scan and replay experiments. **Middle**: 3D trajectories of the end-effector and proxy in the robot base frame during both experiments. **Bottom**: 2D projection of the same trajectories onto the ($x$, $z$) plane, overlaid on a background image of the phantom.

ultrasound image quality. In both the manual and autonomous conditions, ultrasound images were recorded for subsequent comparison of confidence map outputs.

*5.6.2. Results.* In the top left plot of Figure 23, the desired end-effector force trajectory in the transducer frame varies over time due to manual specification by the operator. Despite this variability, the robot successfully tracks the desired force with an average error of 0.30 N. In contrast, the replay experiment uses a fixed desired force of 13.4 N, resulting in a lower average error of 0.18 N.

Regarding moment regulation, the manual scan does not include moment control, leading to higher deviations in the $x$ and $y$ directions. The standard deviation of the sensed moments in the $x$ and $y$ directions during the manual scan is 0.16 N·m and 0.04 N·m, respectively, compared to 0.02 N·m and 0.03 N·m in the replay scan. The average sensed moments in these directions are $-0.08$ N·m and 0.05 N·m, respectively, during the manual scan, compared to $-0.01$ N·m and 0.03 N·m in the replay experiment in which moment control is active. This difference in control strategy also impacts image quality: the average confidence score during the replay experiment is 0.32, which is an 18.5 % improvement over the average confidence of 0.27 in the manual scan. This reflects improved alignment of the transducer with the surface normal during the replay scan.

The robot maintains accurate tracking of the proxy in both cases, with an average positional error of 0.77 cm in the manual scan and 0.45 cm in the replay. The average rotational error for the manual and replay scans are 2.11° in the manual scan and 2.13°, respectively.

In the 3D visualizations, black arrows are plotted every 15 data points to indicate the direction and magnitude of the applied force. The arrow direction reflects the end-effector orientation in the phantom frame, while the arrow length is proportional to the sensed force in the transducer frame. In the manual scan, the arrows exhibit variable lengths and non-uniform orientations, indicating inconsistent force application and alignment. In contrast, the replay scan shows more uniform arrows, demonstrating improved control of both force and moment, and consequently, enhanced ultrasound image quality.

## 6. Conclusion

This work presents an explicit force-based control framework for robotic ultrasound imaging, addressing the fundamental limitations of implicit force regulation in dynamic clinical environments. By utilizing force/torque feedback and controlling force at the end-effector, the proposed system eliminates the sluggish response of intermediate motion loops. Our evaluation demonstrates that this architecture provides a high-bandwidth solution for maintaining precise contact forces and optimal probe orientation, even during dynamic scanning and under external disturbances. We highlight the following results:

- The explicit contact force controller tracked a time-varying force profile representative of physiological motion with less phase lag and greater accuracy than a state-of-the-art implicit force control method. The explicit method also demonstrated greater accuracy and disturbance rejection when tracking a constant desired force with a moving phantom.
- In static scans, force control alone achieved consistent contact, but proper probe alignment and improved image quality were only obtained with the addition of moment control. This alignment effect was consistent across anatomical targets and most significant in soft-tissue regions.
- During dynamic scans over regions of varying stiffness, the system maintained both desired contact force and optimal force-based orientation throughout motion, enabling continuous image acquisition.
- Under external disturbances, the robot compliantly translated and rotated with the phantom while preserving the desired contact force and optimal probe orientation, showing force and motion awareness.
- In a haptically-guided scan, the system successfully followed human-defined trajectories while autonomously regulating constant force and normal orientation, resulting in improved image quality over a manual haptic scan (without assisted constant force and normal orientation control).

A key finding of this study is that while the explicit force controller demonstrates the ability to accurately track dynamic force profiles, as evidenced by the benchmarking and haptically-guided experiments, the determination of the optimal force for a specific diagnostic task requires higher-level reasoning. In this work, we utilized confidence maps as a measure of acoustic coupling to optimize probe orientation; however, this metric characterizes the physics of signal transmission rather than diagnostic utility. Consequently, future research will prioritize the integration of diagnostically-aware objectives, perhaps utilizing learning-based models to generate image-based desired forces in real time (Zakeri et al., 2025).

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## Supplemental material

Supplemental material for this article is available online.

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